

Spectral Unforgetting: Post-Hoc Recovery of Damaged Capabilities Without Retraining

Aarash Abro
Zeta Labs
aarash@zetasolutions.org

Muhammad Tahir
Lahore University of Management Sciences
tahir@lums.edu.pk

Abstract

Fine-tuning a language model for a target task routinely degrades capabilities the training data never explicitly threatened. We study this phenomenon, known as catastrophic forgetting, and propose a post-hoc repair solution that uses only the pretrained checkpoint W_{base} and its fine-tuned descendant W_{ft} . The goal is not merely to revert the model toward the base checkpoint, but to recover capabilities damaged by fine-tuning while preserving both the target-task gains and any beneficial held-out improvements. We introduce DG-Hard, a checkpoint-only spectral repair method for the fine-tuning update $\Delta = W_{\text{ft}} - W_{\text{base}}$. DG-Hard treats Δ as a low-rank task-aligned signal embedded in an IID-like noise residual that gradient descent has no incentive to remove, and applies the Donoho-Gavish hard singular-value threshold to each weight-delta matrix, keeping the structured high-energy part of the update and removing the spectral bulk. This reduces repair to a closed-form SVD filtering step requiring no data-dependent tuning. A central difficulty is evaluation: average accuracy hides per-benchmark failures, while naive recovery scores reward models that simply revert toward the base. We therefore introduce a partition-conditional metric that separately tracks healing, preservation, non-damage, and target-task retention. Across 14 (model, task) settings and nine cross-domain held-out benchmarks, DG-Hard achieves the strongest balanced repair among post-hoc baselines. DG-Hard also restores safety alignment degraded by benign fine-tuning on three independent safety axes, despite using no alignment data. These results suggest that part of fine-tuning-induced capability loss is not an unavoidable consequence of specialization, but a removable spectral residue in the weight update itself. The code can be found at: <https://github.com/BrickleRex/dghard>.

1 Introduction

Fine-tuning can improve a model on the task it is trained for while destroying capabilities that were already present in the pretrained checkpoint. In our experiments, fine-tuning Qwen3.5-4B on the medical domain raises medical-question-answering accuracy, but reduces math-reasoning accuracy drastically. This is not an isolated failure mode: across the 14 (model, task) fine-tuning cells we evaluate, 13 exhibit at least one single-benchmark collapse (Tab. 9, App. E). Such failures are the modern large-model form of *catastrophic forgetting* [1]: adaptation to a new objective can overwrite, distort, or suppress behavior the base model had already acquired [2, 3].

Fine-tuning is the standard way pretrained models are specialized [4, 5, 6, 7], yet its objective contains no term requiring unrelated capabilities to be preserved. The consequences are documented across general-knowledge accuracy [8], safety alignment in aligned models [9], and the geometric distortion of pretrained features outside the training-data span [10]. The checkpoint moves in two ways at once: a structured update lowering the target-task loss, and a residual from many mini-batch SGD steps whose noise scale depends on batch size [11, 12]. The loss rewards only the first; the second

accumulates in directions important for other capabilities, leaving the fine-tuned checkpoint a mixture of task-aligned signal and collateral change.

We study the post-hoc repair of forgetting that has already happened. Given only a base checkpoint W_{base} and a fine-tuned checkpoint W_{ft} , the goal is to recover damaged held-out capabilities while preserving the gains fine-tuning was meant to produce, and has incidentally produced. Reversion toward the base recovers forgotten behavior cheaply but also removes the task-aligned update; a useful repair must distinguish the part of the fine-tuning delta $\Delta = W_{\text{ft}} - W_{\text{base}}$ that carries the new task from the part that causes collateral damage.

Existing post-hoc methods make this decision in coordinate space, whether by scalar interpolation (WiSE-FT, 13), random dropping (DARE, 14), magnitude- and sign-aware pruning (TIES, 15), or forgetting-aware pruning (FAPM, 16). All face the same representation problem: task-relevant entries and harmful residual entries are interleaved by magnitude, sign, and position.

The same update is more separable in singular-value space. Across fine-tuning deltas, the spectrum splits into a broad bulk that matches the random-matrix prediction [17] and a smaller number of outlying spikes that carry the task-aligned update. Both halves are independently supported in prior work: fine-tuning weight updates are rank-deficient [18, 19], and trained-weight spectra fit the Marchenko-Pastur bulk past a finite number of outliers [20, 21]; we verify both on our own deltas in App. A (formalized in §3). Repair becomes a matrix-denoising problem: keep the structure, revert the bulk.

We instantiate this with **DG-Hard**, a closed-form spectral repair: each fine-tuning delta matrix’s SVD is hard-thresholded at the Donoho-Gavish cut [22], and the surviving singular components form a spectrally pruned delta Δ^* that yields the repaired checkpoint $W^* = W_{\text{base}} + \Delta^*$ (Alg. 1). The method is data-free, gradient-free, training-free, and runs in minutes on a single GPU.

Contributions.

1. We formulate post-hoc repair of catastrophic forgetting as a recovery-preservation problem: recover damaged held-out capabilities from a fine-tuned checkpoint while preserving target-task gains and incidental held-out improvements.
2. We identify a spectral structure in fine-tuning deltas. Empirically, damaging residuals concentrate in the singular-value bulk, while task-relevant updates appear as singular-value spikes (App. A). This explains why scalar interpolation and coordinate-wise pruning face an unfavorable trade-off. We propose **DG-Hard**, a data-free repair method that applies the Donoho-Gavish hard threshold [22] to each delta matrix and reconstructs the checkpoint from retained singular components.
3. We introduce a partition-conditional evaluation that reports recovery on damaged measurements, preservation on improved measurements, and retention on unchanged and target-task measurements. DG-Hard achieves the best recovery-preservation trade-off among post-hoc baselines across 14 (model, task) cells and 9 held-out benchmarks (Tab. 1, Fig. 1).

2 Background and related work

Catastrophic forgetting. We extend the connectionist account of McCloskey and Cohen [1] that distributed representations make any helpful update perturb the weights encoding prior tasks [2, 3]. Kumar et al. [10] formalize the modern instance in an overparameterized linear setting, and Luo et al. [8], Qi et al. [9] document its empirical reproduction in contemporary LLMs across general-knowledge and safety benchmarks.

Training-time approaches. Training-time methods modify the optimization step itself and fall into three subfamilies: parameter-movement regularizers, gradient-masking via per-unit importance, and replay against stored past-task data. Parameter-movement regularizers penalize updates weighted by per-parameter importance and differ mostly in how importance is estimated: **EWC** [24] uses the Fisher diagonal, **SI** [25] a path-integral measure of online contribution to the previous-task loss, and **MAS** [26] the gradient of the squared L2 norm of the network output (label-free, since it estimates importance from unlabeled calibration data). We additionally compare against a generic **L1-reg**

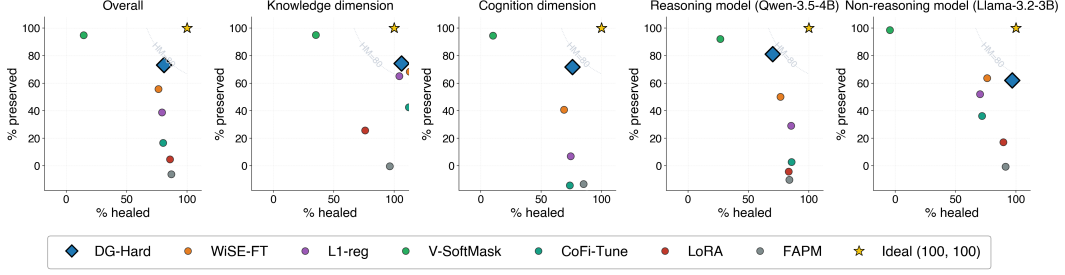


Figure 1: Recovery $\times$ preservation per cohort. Each panel plots the % healed score on the damaged partition (x-axis) against the % preserved score on the improved partition (y-axis), as defined in §4.3. The ideal corner is (100, 100), and the dotted contour marks $HM(\% \text{ healed}, \% \text{ preserved}) = 80$. DG-Hard (blue diamond) is closest to the ideal corner across all five cohorts. FAPM [16] strongly recovers damaged measurements but sacrifices improved ones; V-SoftMask [23] preserves improved measurements but recovers less damage.

baseline that penalizes $\|W - W_{\text{base}}\|_1$ without task information. Gradient-masking methods identify which units carry pretrained capability and slow the gradient flow through them: **DAS** [23] multiplies each gradient by $(1 - \text{importance})$ where importance comes from a dropout-KL proxy on calibration data, and **CoFiTune** [27] extends it with a coarse-grained layer-range filter (restricting the procedure to an empirically selected slice of layers) and a KL/dropout-robustness fine score. Replay-based methods mix or project against stored past-task examples (**GEM** [28], **Experience Replay** [29]) but assume access to a representative pretraining sample, which is unavailable for modern LLMs whose pretraining corpora are proprietary and at terabyte scale.

Parameter-efficient fine-tuning. **LoRA** [18] constrains the fine-tuning update on each weight matrix to a low-rank factorization BA with $r \ll \min(m, n)$, leaving the base weights frozen; the deployed model still merges base + LoRA at inference, so forgetting can persist in the merged weights. Shuttleworth et al. [30] show via direct spectral comparison that this low-rank constraint forces LoRA-FT updates to introduce “intruder” singular directions approximately orthogonal to the pretrained spectrum, and that these intruder directions causally drive forgetting (verified by post-hoc intervention on their singular values).

Post-hoc model merging. The family closest to our work operates post-hoc on $\Delta = W_{\text{ft}} - W_{\text{base}}$ via cheap, data-free transformations. These methods originally targeted multi-task model merging, where multiple fine-tuned checkpoints are combined into one; with a single fine-tune their multi-task aggregation steps (sign election, mean over task vectors) reduce to identity, leaving the per-vector preprocessing as a post-hoc repair on the $(W_{\text{base}}, W_{\text{ft}})$ pair. **WiSE-FT** [13] interpolates linearly between the checkpoints, $W^* = (1-\alpha)W_{\text{base}} + \alpha W_{\text{ft}}$, trading pretrained and fine-tuned behavior with a single scalar. **Task Arithmetic** [31] introduces Δ as a task vector and studies algebraic operations on it (negation, addition, analogies) across multiple tasks. **TIES-Merging** [15] trims low-magnitude entries of each task vector, elects a per-parameter consensus sign across vectors, and averages only the entries aligned with that sign, addressing redundancy and sign disagreement as sources of merge interference. **DARE** [14] randomly drops Δ entries with probability p and rescales survivors by $1/(1-p)$, designed as a preprocessing step that sparsifies fine-tuning deltas before merging. **FAPM** [16] scores each entry with a forgetting-aware criterion combining absolute change magnitude and a relative-change penalty against the pretrained weight, then prunes low-scoring entries. **Fisher-weighted averaging** [32] weights each parameter by its empirical Fisher importance, additionally requiring a small labeled training sample. Per-method hyperparameters are in App. H.1.

Spectral shrinkage and our distinction. The spectral framing fits fine-tuning because the weight update is empirically rank-deficient (18; broader evidence in App. A), and random-matrix theory supplies the matching denoiser. The Marchenko-Pastur theorem [17] fixes the asymptotic spectrum of an IID-noise matrix, and the Donoho-Gavish hard threshold [22] identifies the AMSE-optimal cut between signal and noise (33 extend to optimal continuous shrinkers). Where prior post-hoc methods operate in coordinate space (TIES, DARE, FAPM), as a single scalar (WiSE-FT, Task Arithmetic),

or via labeled-data Fisher weighting, we operate in singular-value space and apply this statistically optimal hard threshold, to our knowledge the first such application to fine-tuning weight deltas for post-hoc forgetting repair.

3 Method

We propose **DG-Hard**: a post-hoc, data-free, gradient-free repair that strips the IID noise residual from a fine-tuning update by applying the optimal hard singular-value threshold of Gavish and Donoho [22] to its SVD. The procedure takes only the base checkpoint W_{base} and the fine-tuned checkpoint W_{ft} , operates one 2D weight matrix at a time, and reduces to a closed-form threshold in the matrix shape and a single estimated noise scale.

3.1 Signal-plus-noise model of the fine-tuning update

For a single 2D weight matrix in $\mathbb{R}^{m \times n}$, write the fine-tuning update as $\Delta = W_{\text{ft}} - W_{\text{base}}$ and take its singular value decomposition,

$$\Delta = U \text{diag}(\mathbf{s}) V^\top, \quad s_1 \geq \dots \geq s_p \geq 0, \quad p = \min(m, n). \quad (1)$$

We model Δ as a two-component sum,

$$\Delta = \Delta_{\text{signal}} + \Delta_{\text{noise}}, \quad (2)$$

with Δ_{signal} a low-rank task-aligned update and Δ_{noise} an IID residual of unknown scale σ . The SGD-noise mechanism in §1 produces exactly this structure: thousands of mini-batch residuals accumulate into the weights while the gradient has no incentive to remove them. Both components are empirically supported: the fine-tuning weight update is rank-deficient [18] and fine-tuning more broadly has low intrinsic dimensionality [19]; trained-weight spectra fit the Marchenko-Pastur bulk past a finite number of outliers [20, 21]. A direct check on our own deltas confirms both properties layer-locally (App. A, Fig. 4). Our goal is to remove Δ_{noise} while leaving Δ_{signal} intact.

3.2 Random-matrix theory and the bulk edge

If Δ were pure IID noise of variance σ^2 , the Marchenko-Pastur theorem [17] pins its spectrum exactly: in the limit $m, n \rightarrow \infty$ at fixed aspect ratio $\beta = \min(m, n) / \max(m, n)$, every singular value sits below the bulk edge

$$\lambda_{\text{MP}} = \sigma (1 + \sqrt{\beta}) \quad (3)$$

with probability one, depending only on σ and β . Contrapositively, any $s_i > \lambda_{\text{MP}}$ in Δ must reflect the signal component of (2). The edge is a principled boundary but not a denoising rule; for that we need a *shrinker* mapping each s_i to a denoised s'_i . The simplest principled choice is a hard threshold, and Gavish and Donoho [22] derive the AMSE-optimal one. Asymptotic mean squared error (AMSE) is the $m, n \rightarrow \infty$ limit of the expected reconstruction error $\mathbb{E}[\|\hat{\Delta}_{\text{signal}} - \Delta_{\text{signal}}\|_F^2]$ (squared Frobenius distance between the estimated and true low-rank signal) at fixed aspect ratio β . A low threshold admits noise (high variance); a high threshold discards signal (high bias); the AMSE-optimal threshold sits at the trade-off minimum.

3.3 DG hard: the optimal hard threshold

Among all hard shrinkers $\eta(s) = s \cdot \mathbf{1}\{s > \tau\}$, Gavish and Donoho [22] identify the threshold minimizing the AMSE of the reconstruction under (2):

$$\tau^* = \omega(\beta) \sigma \sqrt{\max(m, n)}, \quad \omega(\beta) = \sqrt{2(\beta + 1) + \frac{8\beta}{(\beta + 1) + \sqrt{\beta^2 + 14\beta + 1}}}. \quad (4)$$

For square matrices, $\omega(1) = 4/\sqrt{3} \approx 2.309$, the constant for which the original paper is named. The DG-Hard shrinker is

$$\eta_{\text{DG}}(s) = s \cdot \mathbf{1}\{s > \tau^*\}. \quad (5)$$

τ^* sits strictly above λ_{MP} ($\approx 2.309 \sigma \sqrt{n}$ vs. $2\sigma \sqrt{n}$ at $\beta = 1$): cutting at the edge would admit singular values barely distinguishable from noise, paying full reconstruction variance for negligible

Algorithm 1 DG hard repair (per weight matrix).

Require: Base weight $W_{\text{base}} \in \mathbb{R}^{m \times n}$, fine-tuned weight $W_{\text{ft}} \in \mathbb{R}^{m \times n}$.

Ensure: Repaired weight $W^* \in \mathbb{R}^{m \times n}$.

- 1: $\Delta \leftarrow W_{\text{ft}} - W_{\text{base}}$
 - 2: $(U, \mathbf{s}, V) \leftarrow \text{SVD}(\Delta)$
 - 3: $p \leftarrow \min(m, n)$; $\beta \leftarrow p / \max(m, n)$
 - 4: $\hat{\sigma} \leftarrow \text{median}(\mathbf{s}) / (\mu_\beta \sqrt{\max(m, n)})$ ▷ noise scale, App. B
 - 5: $\tau \leftarrow \omega(\beta) \hat{\sigma} \sqrt{\max(m, n)}$ ▷ optimal threshold, (4)
 - 6: $s'_i \leftarrow s_i \cdot \mathbf{1}\{s_i > \tau\}$ **for** $i = 1, \dots, p$
 - 7: $\Delta^* \leftarrow U \text{diag}(\mathbf{s}') V^\top$
 - 8: **return** $W_{\text{base}} + \Delta^*$
-

signal. Eq. (4) is the unique threshold at which marginal bias and variance balance. Algorithm 1 gives the full per-matrix procedure. The noise estimator $\hat{\sigma}$ on line 4 is the median-based Donoho-Gavish estimator $\hat{\sigma} = \text{median}(\mathbf{s}) / (\mu_\beta \sqrt{\max(m, n)})$, which substitutes into (4) to give a data-only threshold $\tau^* = (\omega(\beta) / \mu_\beta) \text{median}(\mathbf{s}) \approx 2.858 \text{median}(\mathbf{s})$ for square matrices; full derivation and the SVD-dominated cost ($O(\min(m, n)^2 \max(m, n))$ per matrix) are in App. B.

3.4 Evaluation protocol

For each held-out benchmark b in every (model, task, method) cell we read $\text{score}_b \in [0, 1]$ and define

$$\Delta_{\text{FT}}(b) = \text{score}_{\text{FT}}(b) - \text{score}_{\text{base}}(b), \quad \Delta_{\text{method}}(b) = \text{score}_{\text{method}}(b) - \text{score}_{\text{base}}(b). \quad (6)$$

At a ± 3 pp threshold each (model, task, b) triple is **damaged** if $\Delta_{\text{FT}}(b) \leq -3$, **improved** if $\Delta_{\text{FT}}(b) \geq +3$, and **unchanged** otherwise. The partition is fixed by the FT checkpoint and identical across methods, so per-method comparisons operate on the same set of triples in each partition cell. Across the 126-cell matrix (2 models $\times$ 7 tasks $\times$ 9 benchmarks) we observe 30 damaged, 55 improved, 41 unchanged triples.

Per-method statistics. Let $s_M = \text{score}_{\text{method}}$, $s_F = \text{score}_{\text{FT}}$, and $s_B = \text{score}_{\text{base}}$. Let D, I, U denote the damaged, improved, and unchanged partition sets, and HM the harmonic mean. Each method produces seven percentage scores. Higher is better; 100 marks full recovery or preservation, values above 100 indicate overshoot, values below 0 indicate regression, and HM aggregates are floored at 0:

$$\% \text{ healed} = \text{avg}_{b \in D} \frac{s_M(b) - s_F(b)}{s_B(b) - s_F(b)} \cdot 100, \quad \% \text{ preserved} = \text{avg}_{b \in I} \frac{s_M(b) - s_B(b)}{s_F(b) - s_B(b)} \cdot 100, \quad (7a)$$

$$\text{on-task ret.} = \text{avg}_{\text{cells}} \frac{s_M(\text{task})}{s_F(\text{task})} \cdot 100, \quad \text{non-damage} = \text{avg}_{b \in U} \mathbf{1}\{s_B(b) - s_M(b) < 3\} \cdot 100, \quad (7b)$$

$$\text{Clean-up} = \text{HM}(\% \text{ healed}, \text{non-damage}), \quad \text{Retention} = \text{HM}(\% \text{ preserved}, \text{on-task ret.}), \quad (7c)$$

$$\text{Combined} = \text{HM}(\text{Clean-up}, \text{Retention}). \quad (7d)$$

Each statistic measures one axis: **% healed** (FT damage recovered on D), **% preserved** (FT gain retained on I), **on-task ret.** (on-task accuracy held vs. FT), **non-damage** (unchanged triples kept within 3 pp of base), and **Clean-up/Retention** (HM of the corresponding pair). “Combined” is the ranking statistic and acts as a triple-bottleneck harmonic mean: collapse on any sub-statistic drives it toward 0, demoting methods that win one axis by sacrificing another (V-SoftMask preserves but does not heal; FAPM heals but does not preserve). Per-cell scores substitute a mean-ratio variant; both aggregations and a comparison to flat unified scores [27] are in App. C.

4 Experiments

A *cell* is one (model, task, held-out benchmark) triple, and a (*model*, *task*) *cell* is one of the 14 fine-tuned checkpoints, each expanding into 9 benchmark cells for a 126-cell matrix. The headline finding

Table 1: Population-level Clean-up, Retention, and Combined scores per method. DG-Hard is the only method scoring ≥ 80 on both Clean-up and Retention; WiSE-FT [13] comes closest but falls to 71.0 on Retention. Every other baseline collapses on at least one axis.

Method	% healed $\uparrow$	Non-damage $\uparrow$	% preserved $\uparrow$	On-task ret. $\uparrow$	Clean-up $\uparrow$	Retention $\uparrow$	Combined $\uparrow$
DG-Hard	80.9 $\pm$ 5.0	85.4 $\pm$ 2.0	73.1 $\pm$ 2.5	97.3 $\pm$ 0.2	83.1 $\pm$ 2.8	83.5 $\pm$ 1.6	83.3
WiSE-FT	76.3 $\pm$ 4.4	90.2 $\pm$ 1.7	55.8 $\pm$ 2.7	97.5 $\pm$ 0.2	82.7 $\pm$ 2.7	71.0 $\pm$ 2.2	76.4
L1-reg	79.3 $\pm$ 4.1	87.8 $\pm$ 2.0	38.7 $\pm$ 3.3	98.6 $\pm$ 0.2	83.4 $\pm$ 2.6	55.6 $\pm$ 3.5	66.7
CoFi-Tune	80.2 $\pm$ 4.8	75.6 $\pm$ 2.2	16.7 $\pm$ 4.3	96.2 $\pm$ 0.2	77.8 $\pm$ 2.7	28.5 $\pm$ 6.2	41.7
V-SoftMask	14.3 $\pm$ 4.5	97.6$\pm$1.9	94.7$\pm$2.3	99.6$\pm$0.2	24.9 $\pm$ 5.9	97.1$\pm$1.2	39.7
LoRA	85.8 $\pm$ 4.8	85.4 $\pm$ 2.0	4.7 $\pm$ 3.3	94.9 $\pm$ 0.2	85.6 $\pm$ 2.7	9.0 $\pm$ 3.5	16.3
FAPM	86.9$\pm$4.4	97.6 $\pm$ 1.7	-6.2 $\pm$ 3.9	77.8 $\pm$ 0.2	91.9$\pm$2.7	0.0 $\pm$ 0.3	0.0

across this matrix: DG-Hard sits at the highest balanced point of the recovery-preservation trade-off, dominating on the reasoning model and matching the strongest baseline on the non-reasoning model. Full per-method per-cell scores are in App. H.

4.1 Setup at a glance

Two models span the reasoning / non-reasoning split: **Qwen3.5-4B** (run in thinking mode with `<think>` blocks stripped at scoring) and **Llama-3.2-3B-Instruct**. Each is fine-tuned under a uniform full-parameter SFT configuration on seven tasks (RTE, StrategyQA, ReClor, BoolQ, MedQA, WikiQA, Winogrande) and evaluated on nine cross-domain held-out benchmarks, partitioned into a 3-benchmark *Knowledge* cohort (MMLU, TriviaQA, TruthfulQA) and a 6-benchmark *Cognition* cohort (ARC-Challenge, GSM8K, IFEval, Math-500, MNLI, HellaSwag). The nine compared methods fall into three groups: reference points **Base** (post-alignment, unfine-tuned) and **Full-SFT** (unrepaired fine-tuned); training-time interventions **L1-reg**, **V-SoftMask** (DAS, 23), **CoFi-Tune** [27], and **LoRA** [18] (rank 16); and post-hoc methods **WiSE-FT** [13], **FAPM** [16], and our **DG-Hard**, the only spectral method in the set; element-wise merging baselines (TIES [15], DARE [14], Task Arithmetic [31]) appear in App. I.3. Inference is identical across methods (vLLM, greedy decoding, no `max_tokens` cap), with a measured ± 1 pp drift floor and a 3 pp significance threshold ($\approx 3\times$ the drift floor) applied throughout. HuggingFace identifiers, eval splits, hyperparameters, per-model repair scope, method configurations, and inference details are in App. D, H.1, and H.2.

4.2 Per-cell forgetting and repair winners

Fine-tuning damages at least one held-out benchmark in 13 of 14 cells, often catastrophically (Qwen/MedQA’s GSM8K collapses from 93.0% to 1.1%). Damage concentrates on a small number of benchmarks per cell rather than spreading uniformly, and the matrix contains 55 improved triples against only 30 damaged ones, so a cell-level mean held-out score smooths the catastrophic drops away and is the wrong instrument to detect forgetting. The rest of the section therefore evaluates methods at the per-benchmark level (per-cell breakdown in Tab. 9, App. E).

A per-cell balance score, HM of the method’s mean held-out ratio against base and its on-task ratio against Full-SFT ($\times 100$), resolves the per-cell winners. **DG-Hard wins 6 of 7 Qwen cells (plus 1 tie); WiSE-FT wins 4 of 7 Llama cells (plus 2 ties); on the 14-cell union, DG-Hard 7, WiSE-FT 4, ties 3.** The reasoning-trained model is more sensitive to the Cognition-side preservation gap that DG-Hard’s spectral threshold widens against the linear average (§4.6); per-cell scores and full discussion in Tab. 10 (App. E).

4.3 Recovery vs preservation (population-level)

Recovery and preservation are coupled: reverting W_{ft} toward W_{base} trivially maximizes recovery but discards the held-out and on-task gains fine-tuning produced. Tab. 1 reports the population-level Clean-up, Retention, and Combined scores (Eq. 7, §3.4) for every method.

Per-method profiles. **DG-Hard** is the only method scoring ≥ 80 on both axes (Combined 83.3). **FAPM** (91.9/0.0) and **V-SoftMask** (24.9/97.1) are the one-axis extremes: FAPM’s 90% sparsity reversion drags improved benchmarks -6.2% below base, while V-SoftMask’s gradient damping preserves the FT distribution but does not actively heal. **LoRA** (85.6/9.0) cleans up well but has no

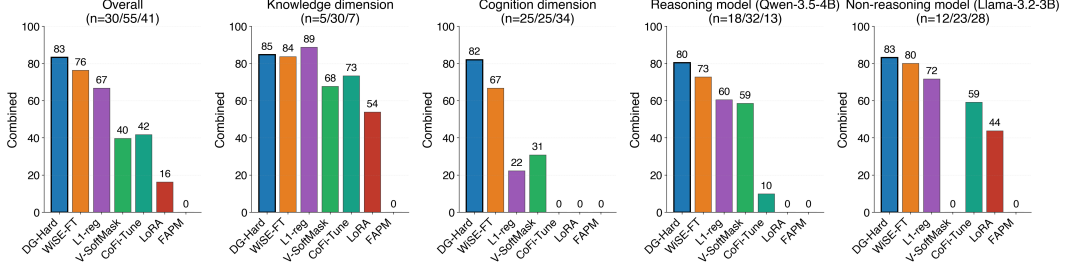


Figure 2: Population-level Combined score per method, sliced by cohort. Panel titles list $n = (\text{damaged/improved/unchanged})$ triple counts per cohort. DG-Hard tops Overall, Cognition, Reasoning, and Non-reasoning; L1-reg edges past DG-Hard on the small- n Knowledge cohort, where its 102.0 Clean-up reflects % healed overshooting base on the 5-case damaged partition. The per-cell balance view in Tab. 10 resolves the Llama cohort into split wins between DG-Hard and WiSE-FT [13]. Methods that collapse on either Clean-up or Retention drop to a low Combined via the harmonic mean’s bottlenecking property.

Table 2: Improvement-bucket % preserved (% , higher is better). Buckets sum to the 55 (model, task, benchmark) triples in the 126-cell matrix.

Bucket	n	DG-Hard	WiSE-FT
3-10 (mild)	25	+65.8 $\pm$ 5.4	+45.3 $\pm$ 5.8
10-20 (moderate)	18	+78.4 $\pm$ 0.9	+70.6 $\pm$ 0.9
20-40 (large)	12	+80.4 $\pm$ 0.5	+55.4 $\pm$ 0.6

Table 3: Damage-bucket % healed (% , higher is better). Buckets sum to the 30 damaged (model, task, benchmark) triples in the 126-cell matrix.

Bucket	n	DG-Hard	WiSE-FT
3-10 (mild)	19	+74.6 $\pm$ 7.0	+65.1 $\pm$ 6.3
10-20 (moderate)	7	+94.6 $\pm$ 1.7	+95.1 $\pm$ 1.8
20-40 (severe)	3	+86.3 $\pm$ 1.7	+98.1 $\pm$ 1.7
40+ (extreme)	1	+87.6 $\pm$ 0.4	+92.3 $\pm$ 0.4

FT-side gain to preserve because low-rank adapters never produced FT’s incidental held-out gains. CoFi-Tune, L1-reg, and WiSE-FT are middling on both.

4.4 Cohort breakdown

DG-Hard wins Combined on four of five cohorts (Overall, Cognition, Reasoning, Non-reasoning) by maintaining strong Clean-up and strong Retention simultaneously (Fig. 2; full per-cohort Clean-up and Retention in App. F). On the small- n Knowledge cohort (5 damaged triples) L1-reg overshoots base in healing and edges past on Combined; this advantage does not transfer to the larger Cognition cohort, where L1-reg’s Retention collapses to 13.0 and its Combined falls to 22.3. **DG-Hard’s edge over WiSE-FT comes mostly from Retention:** Clean-up scores are within ~ 3 pp on most cohorts, but Retention is 12+ pp higher on Overall, Cognition, and Reasoning, driven by held-out preservation since on-task retention is essentially flat at $\sim 97\%$ for both.

4.5 Bucketed view

Bucketing damaged and improved triples by FT-effect magnitude (Tabs. 2, 3; per-cohort sub-score breakdown in App. G) resolves where the DG-Hard / WiSE-FT trade-off is actually decided.

Damage-recovery is roughly tied. Both methods heal most of the damage in every bucket (Tab. 3); the trade-off is not decided here.

Preservation widens with FT-gain magnitude. The preservation gap grows with the size of the FT lift (Tab. 2): on the largest lifts WiSE-FT loses about half the gain while DG-Hard keeps four-fifths. This matches the spectral mechanism: DG-Hard’s threshold retains the high-singular-value directions of Δ that a linear weight average uniformly attenuates, so the more concentrated the FT signal, the wider DG-Hard’s preservation lead. The damage side runs the same direction in expectation: the mild bucket holds 19 of the 30 damaged triples and DG-Hard wins it, so even where average healing reads as a tie the dominant mode of forgetting falls to DG-Hard (Tab. 3).

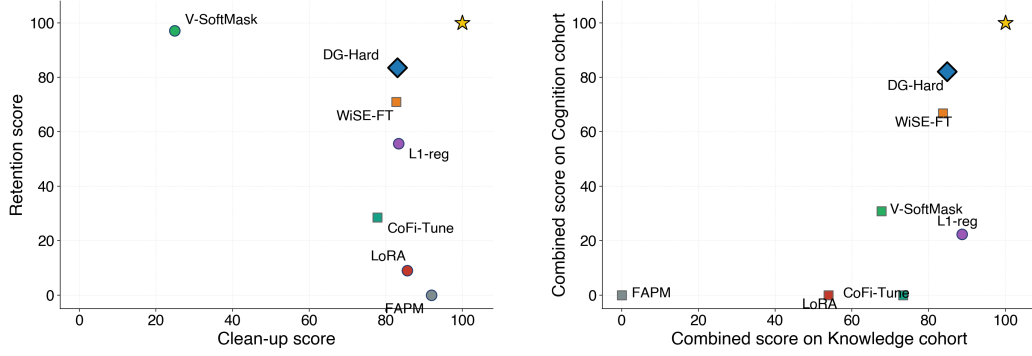


Figure 3: Trade-off frontiers (both axes 0 to 100, higher is better). *Left*: Clean-up vs Retention. DG-Hard sits in the upper-right region where both axes are simultaneously high; V-SoftMask [23] is the retention-extreme (top-left); FAPM [16] is the cleanup-extreme (bottom-right). *Right*: Knowledge-cohort Combined (x) vs Cognition-cohort Combined (y). DG-Hard sits high on both (84.8 and 82.1), the most balanced strong method; WiSE-FT [13] matches on Knowledge but loses ground on Cognition (66.8) because its preservation drops there; V-SoftMask sits in the Knowledge-favouring region (67.7 vs 30.9); CoFi-Tune [27], FAPM, and LoRA [18] collapse on Cognition due to non-positive average preservation.

4.6 Cognition and the trade-off frontier

The Knowledge cohort ($n = 5$ damaged) is too easy to differentiate methods on healing: every method except V-SoftMask recovers at or above base on the damaged Knowledge cases, with several overshooting (per-method Clean-up and Retention per cohort in Tabs. 11, 12, App. F; underlying % healed and % preserved in the corresponding panels of Fig. 1).

The Cognition cohort ($n = 25$ damaged) is where the field separates. Healing remains broadly comparable across non-pathological baselines, but Retention collapses for every method except DG-Hard and V-SoftMask: DG-Hard’s Cognition Retention of 82.5 is roughly 25 pp above the next strongest baseline (WiSE-FT at 57.5) and substantially higher than L1-reg (13.0), while CoFi-Tune, FAPM, and LoRA collapse to zero because their average % preserved on the Cognition partition is non-positive (Tab. 12). **The Cognition preservation gap, is what separates DG-Hard from the rest of the field and is the most plausible mechanism behind its reasoning-model lead.** V-SoftMask preserves Cognition at near-perfect levels but heals almost nothing on the same cohort, so it does not function as a usable repair for any downstream task that requires reasoning.

The same separation appears geometrically in Fig. 3. Only DG-Hard sits in the upper-right of both panels; FAPM and V-SoftMask anchor opposite corners (extreme on one axis at the cost of the other); WiSE-FT trails DG-Hard but stays closest among the baselines; L1-reg, CoFi-Tune, and LoRA spread along the lower edges (low Retention on the left, low Cognition Combined on the right). For repair tasks, where neither axis can be allowed to collapse, this rules out the corners and the lower edges alike.

4.7 Safety alignment recovery

Both base models are safety aligned at the Instruct tier, and benign supervised fine-tuning on knowledge or reasoning data is not designed to touch alignment. The question is whether the FT updates we study erode safety as an incidental side effect, and whether DG-Hard, tuned only on capability metrics, restores it without being asked to. We score each {Base, FT, DG-Hard} triple on three axes: HarmBench v1.0 [34] for refusal of harmful prompts, XSTest v2 [35] for over-refusal of safe prompts, and StrongREJECT v1.0 [36] for continuous-rubric harmfulness on forbidden prompts (Tab. 4).

The signal-plus-noise model predicts that safety alignment, encoded in pretrained directions Base already occupies, should survive a repair that removes only the IID noise residual. The data is consistent with this prediction. On every Llama cell where FT lowered HarmBench refusal or raised

Table 4: Per-cell Base / FT / DG-Hard scores on three safety axes: HarmBench refusal [34] ($\uparrow$), XSTest over-refusal [35] ($\downarrow$), and StrongREJECT harm [36] ($\downarrow$). **Bold** = DG-Hard best among {Base, FT, DG}; underline = second. Eval details in App. D.3.

Model	Task	HarmBench refusal $\uparrow$			XSTest over-refusal $\downarrow$			StrongREJECT harm $\downarrow$		
		Base	FT	DG	Base	FT	DG	Base	FT	DG
qwen3p5_4b	winogrande	96.0	81.0	<u>94.5</u>	2.8	0.8	<u>1.2</u>	0.0160	0.0160	0.0096
qwen3p5_4b	wikiqa	97.0	100.0	<u>97.0</u>	2.8	15.6	<u>4.0</u>	0.0160	0.0076	<u>0.0080</u>
qwen3p5_4b	medqa	95.0	99.5	<u>91.5</u>	2.0	4.4	1.2	0.0076	0.0096	0.0128
llama3p2_3b	winogrande	88.0	75.5	<u>84.0</u>	8.0	3.6	<u>5.6</u>	0.0479	0.1402	<u>0.0931</u>
llama3p2_3b	wikiqa	87.5	84.5	87.5	7.6	4.4	<u>6.8</u>	0.0531	0.0567	0.0463
llama3p2_3b	medqa	87.5	79.0	<u>84.0</u>	7.6	4.0	<u>5.2</u>	0.0539	0.0795	<u>0.0663</u>

StrongREJECT harmfulness, DG-Hard recovers the majority of the gap toward Base. No cell shows large degradation, including the Qwen cells where FT incidentally improved a safety axis.

Qwen+wikiqa provides a more demanding test. FT shifts alignment in opposite directions on the same model: refusal of harmful prompts saturates at 100%, while XSTest over-refusal of safe prompts rises from 2.8 to 15.6. DG-Hard returns both axes to within small residuals of Base. That a non-safety-specific spectral repair simultaneously corrects both opposite-direction shifts on the same cell is consistent with the bidirectional damage sharing the noise-residual structure DG-Hard targets.

StrongREJECT’s continuous rubric provides a finer-grained signal than the binary refusal label: on Llama, non-refusals are accompanied by greater per-occurrence severity, and DG-Hard recovers a substantial fraction of that severity. The threshold was set without reference to alignment data, so the safety recovery appears to follow from the noise-residual model the spectral cut is built around, rather than from a designed objective.

The broader pattern across capability and safety axes is consistent. DG-Hard occupies a balanced position between Base and FT: damaged held-out capabilities recover toward Base, on-task gains track FT, and safety alignment is preserved as an apparent side effect of the same spectral cut. The three axes appear to track one mechanism rather than three.

5 Limitations and conclusion

Limitations. DG-Hard treats the task-relevant update as concentrated in a small number of high-singular-value directions; small-spectral-energy components are zeroed regardless of semantic importance, which can fail when the useful signal is spectrally diffuse. This assumption is consistent with prior work showing that fine-tuning weight updates are rank-deficient [18], that the fine-tuning objective has low intrinsic dimensionality [19], and that full-FT updates carry higher effective rank than LoRA reparameterizations [30] (App. A); the failure mode applies to a minority regime but is not ruled out a priori.

The only operational consideration is the per-matrix SVD cost, $O(\min(m, n)^2 \max(m, n))$. Even at frontier scale the dominant matrix grows only $\sim 7\times$ from Llama-3.2-3B’s embedding ($128,256 \times 3,072$) to Kimi-K2.6’s ($163,840 \times 7,168$). At the ~ 1 T-parameter scale of Kimi-K2.6, the remaining mixture-of-experts components admit parallel SVD; a full repair completes in hours, a negligible fraction of the weeks of training that produced the checkpoints.

Conclusion. Fine-tuning weight deltas admit a clean two-component decomposition: a low-rank task-aligned update and an IID-like noise residual that gradient descent has no incentive to remove. DG-Hard strips the residual by applying the Donoho-Gavish optimal hard SVD threshold post hoc, data-free, and gradient-free. Across 14 (model, task) cells on Qwen3.5-4B and Llama-3.2-3B-Instruct and nine cross-domain held-out benchmarks, DG-Hard achieves the highest balanced point on the recovery-preservation trade-off under a partition-conditional Combined metric, beating the next-best post-hoc baseline by +6.9 pp, and restores safety alignment that benign fine-tuning incidentally erodes on three independent safety axes without alignment data entering the procedure. The capability and safety patterns appear to follow from a single mechanism rather than three. Additional experiments supporting these design choices, including a comparison with element-wise model-merging baselines, are in App. I.

References

- [1] Michael McCloskey and N. J. Cohen. Catastrophic interference in connectionist networks: The sequential learning problem. In *Psychology of Learning and Motivation*, volume 24, pages 109–165. Academic Press, 1989.
- [2] Roger Ratcliff. Connectionist models of recognition memory: Constraints imposed by learning and forgetting functions. *Psychological Review*, 97(2):285–308, 1990. doi: 10.1037/0033-295X.97.2.285.
- [3] Robert M. French. Catastrophic forgetting in connectionist networks. *Trends in Cognitive Sciences*, 3(4):128–135, 1999. doi: 10.1016/S1364-6613(99)01294-2.
- [4] Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel M. Ziegler, Jeffrey Wu, Clemens Winter, Christopher Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. Language models are few-shot learners. In *Advances in Neural Information Processing Systems*, 2020. URL <https://arxiv.org/abs/2005.14165>.
- [5] Jeremy Howard and Sebastian Ruder. Universal language model fine-tuning for text classification. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics*, 2018. URL <https://arxiv.org/abs/1801.06146>.
- [6] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 2019. URL <https://arxiv.org/abs/1810.04805>.
- [7] Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul Christiano, Jan Leike, and Ryan Lowe. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems*, 2022. URL <https://arxiv.org/abs/2203.02155>.
- [8] Yun Luo, Zefan Yang, Fan Meng, Yufei Li, Jie Zhou, and Yue Zhang. An empirical study of catastrophic forgetting in large language models during continual fine-tuning. *arXiv preprint arXiv:2308.08747*, 2023. URL <https://arxiv.org/abs/2308.08747>.
- [9] Xiangyu Qi, Yi Zeng, Tinghao Xie, Pin-Yu Chen, Ruoxi Jia, Prateek Mittal, and Peter Henderson. Fine-tuning aligned language models compromises safety, even when users do not intend to! In *International Conference on Learning Representations*, 2024. URL <https://arxiv.org/abs/2310.03693>.
- [10] Ananya Kumar, Aditi Raghunathan, Robbie Jones, Tengyu Ma, and Percy Liang. Fine-tuning can distort pretrained features and underperform out-of-distribution. In *International Conference on Learning Representations*, 2022.
- [11] Stanisław Jastrzębski, Zachary Kenton, Devansh Arpit, Nicolas Ballas, Asja Fischer, Yoshua Bengio, and Amos Storkey. Three factors influencing minima in SGD. *arXiv preprint arXiv:1711.04623*, 2017.
- [12] N. S. Keskar, Dheevatsa Mudigere, Jorge Nocedal, Misha Smelyanskiy, and Ping Tak Peter Tang. On large-batch training for deep learning: Generalization gap and sharp minima. In *International Conference on Learning Representations*, 2017.
- [13] Mitchell Wortsman, Gabriel Ilharco, Jong Wook Kim, Mike Y. Li, Simon Kornblith, Rebecca Roelofs, Raphael Gontijo Lopes, Hannaneh Hajishirzi, Ali Farhadi, Hongseok Namkoong, and Ludwig Schmidt. Robust fine-tuning of zero-shot models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022. URL <https://arxiv.org/abs/2109.01903>.

- [14] Le Yu, Bowen Yu, Haiyang Yu, Fei Huang, and Yong Li. Language models are super mario: Absorbing abilities from homologous models as a free lunch. In *International Conference on Machine Learning*, 2024. URL <https://arxiv.org/abs/2311.03099>.
- [15] Prateek Yadav, Derek Tam, Leshem Choshen, Colin Raffel, and Mohit Bansal. TIES-merging: Resolving interference when merging models. In *Advances in Neural Information Processing Systems*, 2023. URL <https://arxiv.org/abs/2306.01708>.
- [16] Wei Huang, Aimin Cheng, and Yu Wang. Mitigating catastrophic forgetting in large language models with forgetting-aware pruning (FAPM). In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, 2025. URL <https://arxiv.org/abs/2509.08255>.
- [17] V. A. Marchenko and L. A. Pastur. Distribution of eigenvalues for some sets of random matrices. *Mathematics of the USSR-Sbornik*, 1(4):457–483, 1967.
- [18] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022. URL <https://arxiv.org/abs/2106.09685>.
- [19] Armen Aghajanyan, Luke Zettlemoyer, and Sonal Gupta. Intrinsic dimensionality explains the effectiveness of language model fine-tuning. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics*, 2021. URL <https://arxiv.org/abs/2012.13255>.
- [20] Matthias Thamm, Max Staats, and Bernd Rosenow. Random matrix analysis of deep neural network weight matrices. *Physical Review E*, 106:054124, 2022. URL <https://arxiv.org/abs/2203.14661>.
- [21] Max Staats, Matthias Thamm, and Bernd Rosenow. Boundary between noise and information applied to filtering neural network weight matrices. *Physical Review E*, 108:L022302, 2023. URL <https://arxiv.org/abs/2206.03927>.
- [22] Matan Gavish and David L. Donoho. The optimal hard threshold for singular values is $4/\sqrt{3}$. *IEEE Transactions on Information Theory*, 60(8):5040–5053, 2014. URL <https://arxiv.org/abs/1305.5870>.
- [23] Zixuan Ke, Yijia Shao, Haolong Lin, Tatsuya Konishi, Gyuwan Kim, and Bing Liu. Continual pre-training of language models. In *International Conference on Learning Representations*, 2023. URL <https://arxiv.org/abs/2302.03241>.
- [24] James Kirkpatrick, Razvan Pascanu, Neil Rabinowitz, Joel Veness, Guillaume Desjardins, Andrei A. Rusu, Kieran Milan, John Quan, Tiago Ramalho, Agnieszka Grabska-Barwińska, Demis Hassabis, Claudia Clopath, Dharshan Kumaran, and Raia Hadsell. Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences*, 114(13):3521–3526, 2017. URL <https://arxiv.org/abs/1612.00796>.
- [25] Friedemann Zenke, Ben Poole, and Surya Ganguli. Continual learning through synaptic intelligence. In *International Conference on Machine Learning*, 2017. URL <https://arxiv.org/abs/1703.04200>.
- [26] Rahaf Aljundi, Francesca Babiloni, Mohamed Elhoseiny, Marcus Rohrbach, and Tinne Tuytelaars. Memory aware synapses: Learning what (not) to forget. In *Proceedings of the European Conference on Computer Vision*, 2018. URL <https://arxiv.org/abs/1711.09601>.
- [27] Haonan Zhang, Yinjun Wu, Dongxu Li, Shuo Yang, Rui Zhao, Yu Jiang, and Fei Tan. Balancing speciality and versatility: A coarse to fine framework for mitigating catastrophic forgetting in large language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, 2024. URL <https://arxiv.org/abs/2404.10306>.
- [28] David Lopez-Paz and Marc’Aurelio Ranzato. Gradient episodic memory for continual learning. In *Advances in Neural Information Processing Systems*, 2017. URL <https://arxiv.org/abs/1706.08840>.

- [29] Arslan Chaudhry, Marcus Rohrbach, Mohamed Elhoseiny, Thalaiyasingam Ajanthan, Puneet K. Dokania, Philip H. S. Torr, and Marc’Aurelio Ranzato. On tiny episodic memories in continual learning. *arXiv preprint arXiv:1902.10486*, 2019. URL <https://arxiv.org/abs/1902.10486>.
- [30] Richard Shuttlesworth, Jacob Andreas, Antonio Torralba, and Pratyusha Sharma. LoRA vs full fine-tuning: An illusion of equivalence. In *Advances in Neural Information Processing Systems*, 2025. URL <https://arxiv.org/abs/2410.21228>.
- [31] Gabriel Ilharco, Marco Tulio Ribeiro, Mitchell Wortsman, Suchin Gururangan, Ludwig Schmidt, Hannaneh Hajishirzi, and Ali Farhadi. Editing models with task arithmetic. In *International Conference on Learning Representations*, 2023. URL <https://arxiv.org/abs/2212.04089>.
- [32] Michael S. Matena and Colin A. Raffel. Merging models with fisher-weighted averaging. In *Advances in Neural Information Processing Systems*, 2022. URL <https://arxiv.org/abs/2111.09832>.
- [33] Matan Gavish and David L. Donoho. Optimal shrinkage of singular values. *IEEE Transactions on Information Theory*, 63(4):2137–2152, 2017. URL <https://arxiv.org/abs/1405.7511>.
- [34] Mantas Mazeika, Long Phan, Xuwang Yin, Andy Zou, Zifan Wang, Norman Mu, Elham Sakhae, Nathaniel Li, Steven Basart, Bo Li, David Forsyth, and Dan Hendrycks. HarmBench: A standardized evaluation framework for automated red teaming and robust refusal. In *Proceedings of the 41st International Conference on Machine Learning*, 2024. URL <https://arxiv.org/abs/2402.04249>.
- [35] Paul Röttger, Hannah Rose Kirk, Bertie Vidgen, Giuseppe Attanasio, Federico Bianchi, and Dirk Hovy. XSTest: A test suite for identifying exaggerated safety behaviours in large language models. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics*, 2024. URL <https://arxiv.org/abs/2308.01263>.
- [36] Alexandra Souly, Qingyuan Lu, Dillon Bowen, Tu Trinh, Elvis Hsieh, Sana Pandey, Pieter Abbeel, Justin Svegliato, Scott Emmons, Olivia Watkins, and Sam Toyer. A StrongREJECT for empty jailbreaks. In *Advances in Neural Information Processing Systems (Datasets and Benchmarks Track)*, 2024. URL <https://arxiv.org/abs/2402.10260>.
- [37] Abhishek Panigrahi, Nikunj Saunshi, Haifeng Zhao, and Sanjeev Arora. Task-specific skill localization in fine-tuned language models. In *International Conference on Machine Learning*, 2023. URL <https://arxiv.org/abs/2302.06600>.
- [38] Pratyusha Sharma, Jordan T. Ash, and Dipendra Misra. The truth is in there: Improving reasoning in language models with layer-selective rank reduction. In *International Conference on Learning Representations*, 2024. URL <https://arxiv.org/abs/2312.13558>.

Appendix

Contents

A	Empirical evidence for the signal-plus-noise structure of Δ	14
B	Noise-scale derivation	15
C	Metric discussion	16
D	Detailed experimental setup	17
D.1	Fine-tuning tasks	17
D.2	Held-out benchmarks	17
D.3	Training hyperparameters	18
D.4	Repair scope	18
E	Per-cell forgetting and balance scores	19
F	Per-cohort Clean-up and Retention	20
G	Full per-cohort sub-score breakdown	21
H	Per-(model, task, method, benchmark) results	21
H.1	Method-specific configurations	22
H.2	Inference engine and decoding	22
I	Other experiments and ablations	22
I.1	Layer-wise noise concentration	22
I.2	Layer-mask causal test	25
I.3	Comparison with element-wise merging baselines	26

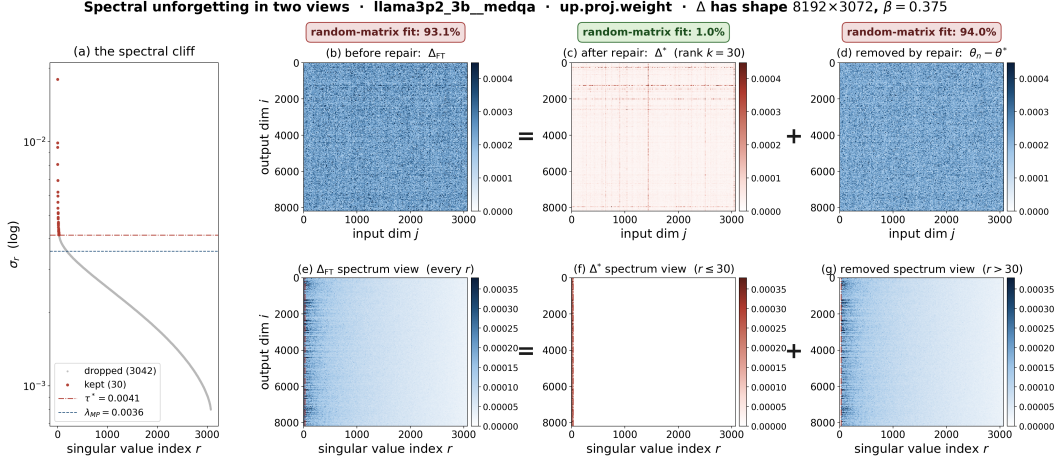


Figure 4: Spectral unforgetting in two views, on Llama-3.2-3B mlp.up_proj at layer 14, with $\Delta = W_{\text{ft}} - W_{\text{base}} \in \mathbb{R}^{8192 \times 3072}$ ($\beta = 0.375$). **(a)** The fine-tune delta has a sharp spectral cliff: 30 singular values (red) lie above the DG-Hard threshold $\tau^* = \omega(\beta)\hat{\sigma}$, while the remaining 3042 (gray) sit at or below the Marchenko-Pastur bulk edge λ_{MP} . **Top row, (b) to (d): entry-space view** of the additive identity $\Delta_{\text{FT}} = \Delta^* + (W_{\text{ft}} - W^*)$, with each pixel showing $\max |\Delta_{ij}|$ over a block of matrix entries. The full FT delta (b) is uniform speckle; the rank-30 repaired delta (c) reveals horizontal banding from the kept left-singular vectors u_r ; the discarded component (d) is again uniform speckle, carrying no spatial structure. **Bottom row, (e) to (g): spectrum-space view** of the same identity, with each pixel showing $\max \sigma_r |u_r[i]|$, the contribution of singular direction r to output neuron i . Panel (e) decomposes into the kept-only panel (f), nonzero only in its leftmost 30 columns, and the bulk-only panel (g), nonzero everywhere except those columns; their pixel-wise sum reproduces (e) exactly, because every singular direction belongs to exactly one set. Random-matrix-fit chips above the top row report the percentage match between each matrix’s singular-value distribution and the Marchenko-Pastur prediction (green: structured, non-random; red: IID-noise-like): Δ_{FT} scores 93.1%, the rank-30 repaired delta Δ^* scores 1.0%, and the discarded component scores 94.0%, slightly more MP-like than Δ_{FT} itself, confirming that DG-Hard separates the two without leaving residual signal in the noise.

A Empirical evidence for the signal-plus-noise structure of Δ

The central modeling assumption of selective spectral reversion is that, for each 2D weight matrix W in the network, the fine-tuning delta $\Delta = W_{\text{ft}} - W_{\text{base}}$ (equivalently ΔW in much of the prior LoRA-adjacent literature) admits a two-component decomposition $\Delta = \Delta_{\text{signal}} + \Delta_{\text{noise}}$, where Δ_{signal} is a low-rank task-aligned update and Δ_{noise} is a residual that behaves statistically like an IID random matrix. The phenomenological evidence the model is designed to explain is the per-cell forgetting pattern in Tab. 9: every fine-tune produces a small set of large per-benchmark drops on a small set of held-out benchmarks (concentrated signal-side damage), while the remaining held-out scores drift only mildly (diffuse noise-side damage). Two independent threads of empirical evidence, one from the fine-tuning literature and one from the random-matrix-theory literature, support this picture, and a direct check on our own checkpoints (Fig. 4) completes the argument.

The signal side: the task-aligned update is low-dimensional. Aghajanyan et al. [19] establish low intrinsic dimensionality for the fine-tuning objective by showing that optimizing only ~ 200 trainable parameters via random projection into the full parameter space recovers 90% of full-fine-tune performance on MRPC with RoBERTa; this is a property of the optimization landscape rather than of the realized weight delta. Hu et al. [18] provide the more direct evidence: in their intrinsic-rank analysis, very low LoRA ranks (often single-digit) match high-rank LoRA on downstream tasks, and the learned ΔW amplifies a small set of task-specific directions, supporting their explicit “rank-deficiency of ΔW ” conclusion. Shuttleworth et al. [30] report that full fine-tuning carries a higher effective rank than LoRA reparameterizations even at matched nominal rank, with full-FT

deltas modifying the existing pretrained singular structure and LoRA introducing a small number of additional “intruder” directions approximately orthogonal to that structure; the spike-versus-bulk reading of our signal-plus-noise model survives both regimes because it isolates the structured component above the MP bulk rather than committing to a strict low-rank cap. Panigrahi et al. [37] provide a complementary parametric-sparsity result: $\sim 0.01\%$ of parameters carry $> 95\%$ of fine-tune task performance when grafted back onto the base model. Sparsity in coordinate space and concentration in singular-value space are distinct mathematical properties; we cite this work as a parallel structural prior, not as direct evidence for low-rank Δ .

The noise side: the residual past the spikes follows the Marchenko-Pastur bulk. Random-matrix theory is an empirically validated description of trained neural-network weight matrices, not just an asymptotic abstraction. Thamm et al. [20] analyze the singular spectra of trained deep-network weight matrices and demonstrate that the bulk fits the MP density layer by layer, with a finite number of outliers carrying the learned signal. Staats et al. [21] operationalize the same picture: they apply MP-edge filtering directly to neural-network weight matrices, set the sub-edge bulk to zero, and recover the trained network’s behavior, providing direct evidence that the bulk is functionally inert. Two independent fine-tuning-side observations point in the same direction. Sharma et al. [38] (LASER) show that aggressively zeroing the high-rank tail of a trained transformer’s weight matrix can improve downstream performance, suggesting the high-rank bulk encodes noise rather than usable signal, a result that transfers to the fine-tune delta by linearity of SVD truncation. Yu et al. [14] (DARE) show that the fine-tune delta tolerates random pruning of 90–99% of its entries with rescaling, attributing this to “extreme redundancy” of small-magnitude updates, consistent with most of Δ being redundant rather than informative.

Direct check on our own checkpoints. Fig. 4 verifies the structure layer-locally on Llama-3.2-3B’s `mlp.up.proj` at layer 14. Panel (a) shows the spectral cliff at the DG-Hard threshold τ^* . Panels (b) to (g) show that the additive identity $\Delta_{\text{FT}} = \Delta^* + (W_{\text{ft}} - W^*)$ holds simultaneously in entry space (top row) and in singular-vector space (bottom row). The random-matrix-fit chips on the top row verify the decomposition quantitatively. We define

$$\text{MP-fit}(\mathbf{s}) = (1 - \text{KS}(F_{\mathbf{s}}, F_{\text{MP}}(\hat{\sigma}, \beta))) \cdot 100, \quad (8)$$

where $F_{\mathbf{s}}$ is the empirical CDF of the matrix’s singular values, F_{MP} is the closed-form Marchenko-Pastur CDF at the noise-scale estimate $\hat{\sigma}$ shared across the three matrices and the aspect ratio β , and KS is the Kolmogorov-Smirnov distance between them; MP-fit = 100 indicates that the empirical and theoretical CDFs coincide ($\mathbf{s}$ is statistically indistinguishable from IID noise), while MP-fit $\rightarrow 0$ indicates a maximally non-random spectrum. Under this metric, Δ_{FT} scores 93.1%, the rank-30 repaired delta Δ^* scores 1.0% (explicitly non-random), and the discarded component scores 94.0%, slightly more MP-like than Δ_{FT} itself, confirming that DG-Hard extracts the structured component without leaving residual signal in the noise.

Where this lands the central assumption. The signal-side and noise-side evidence jointly support the decomposition: the weight update is rank-deficient [18] and the fine-tuning objective has low intrinsic dimensionality [19], with related findings on the spectral structure of full-FT versus LoRA deltas [30] and on parametric sparsity [37]; the residual is empirically MP-bulk-distributed [20, 21], and that bulk is functionally redundant under direct manipulation of trained weights and fine-tune deltas [14, 38]. Fig. 4 checks the bulk-fit and additivity claims on the exact Δ matrix the method operates on.

B Noise-scale derivation

§3.3 of the main paper introduces the Donoho-Gavish noise estimator. This appendix gives the full derivation and the implementation footprint.

Equation (4) requires the noise scale σ of the residual Δ_{noise} in (2). Ideally we would estimate σ from Δ_{noise} directly, but the decomposition in (2) is unobservable. Gavish and Donoho [22] resolve this within the same MP framework. Under the spike model, only the top r singular values carry signal contamination, with r unknown but small relative to p ; the remaining $p - r$ are asymptotically

distributed as the singular-value MP density at aspect β . That density has a closed-form median μ_β ,

$$\mu_\beta = \sqrt{\text{median}(f_\beta)}, \quad f_\beta(\lambda) = \frac{\sqrt{(\lambda_+ - \lambda)(\lambda - \lambda_-)}}{2\pi\beta\lambda}, \quad \lambda_\pm = (1 \pm \sqrt{\beta})^2, \quad (9)$$

which we precompute by fine-grid trapezoidal integration once per distinct aspect ratio in the network. Because $r \ll p$, the median of the *empirical* singular values $\text{median}(\mathbf{s})$ falls inside the noise bulk regardless of the precise value of r , and asymptotically tracks $\sigma\sqrt{\max(m, n)} \cdot \mu_\beta$. Solving for σ gives the Donoho-Gavish noise estimator,

$$\hat{\sigma} = \frac{\text{median}(\mathbf{s})}{\mu_\beta \sqrt{\max(m, n)}}. \quad (10)$$

Equation (10) is internally consistent with the rest of the DG framework: the same MP density that determines the bulk edge (3) and the optimal threshold (4) also determines μ_β . Substituting (10) into (4) eliminates $\sqrt{\max(m, n)}$ and yields the unconditional form of the DG hard threshold,

$$\tau^* = \frac{\omega(\beta)}{\mu_\beta} \text{median}(\mathbf{s}), \quad (11)$$

which evaluates to $\approx 2.858 \cdot \text{median}(\mathbf{s})$ for square matrices and is computable directly from the empirical spectrum without any auxiliary statistic.

Cost and implementation footprint. The per-matrix dominant cost is the SVD on line 4’s preceding step, $O(\min(m, n)^2 \max(m, n))$, completing in a few hundred milliseconds for the largest matrix in the model sizes we evaluate (the Llama-3.2-3B-Instruct embedding, $128,256 \times 3,072$); full-network repair completes in ≤ 3 minutes on a single A100-80GB for the model sizes we evaluate. The implementation loads W_{base} and W_{ft} once, processes Algorithm 1 tensor-by-tensor, and stitches repaired matrices back at their original keys. The repair scope is restricted to parameters with $\text{ndim} \geq 2$ and at least 1024 elements; tensors with $\text{ndim} > 2$ (e.g., 1D convolution kernels) are reshaped to 2D before SVD and reshaped back afterwards. 1D parameters (biases, normalization scales) are left at their fine-tuned values since they admit no non-trivial MP edge. Peak GPU memory is approximately twice the model’s parameter footprint plus a working buffer for the batched SVD, well within a 40 GB A100.

C Metric discussion

% healed. Conventional held-out accuracy averages across all benchmarks regardless of whether fine-tuning damaged them, so a method that simply mirrors FT scores indistinguishably from one that actively repairs. % healed restricts the average to the FT-damaged partition D and reports how much of the FT-to-Base gap each method closes. The numerator is the repaired score’s gain over FT; the denominator is the gap fine-tuning opened, so 0 means no recovery and 100 means full restoration to Base.

% preserved. The analogous construction on the FT-improved partition I . A flat held-out average pools FT-improved triples with FT-unchanged ones, so a method that erases every incidental gain looks numerically similar to one that keeps them whenever most cells were FT-untouched. % preserved isolates the triples where fine-tuning produced a real lift and reports the share of that lift the repaired model still carries.

Non-damage. Repairs that aggressively shrink the delta can introduce regressions on benchmarks fine-tuning never touched. A flat held-out mean smooths these regressions into the average; non-damage surfaces them. The statistic is the fraction of FT-unchanged triples on which the repaired score is within 3 pp of Base, so a method that bleeds outside the FT-damaged cells loses points here independently of how well it heals.

On-task retention. Absolute on-task accuracy is misleading: tasks vary in difficulty, and a flat report rewards methods that simply leave the FT model unchanged. On-task ret. normalizes the method’s target-task score by FT’s, so 100 means every fine-tuning gain on the target task is preserved and 0 means none is.

Clean-up and Retention. Clean-up = $\text{HM}(\% \text{ healed, non-damage})$ summarizes the cleaning side: actively reverting FT-damage on the FT-damaged set D and leaving the FT-unchanged set U alone. Retention = $\text{HM}(\% \text{ preserved, on-task ret.})$ summarizes the preservation side: keeping incidental held-out gains on the FT-improved set I and the target-task gain. Each is a harmonic mean by design, so a method that maximizes one sub-statistic by collapsing the other is bottlenecked rather than rewarded; Combined applies the same bottleneck across the two sides.

Why not a flat unified average? A common alternative aggregation in the continual-learning literature, the Uni. score of Zhang et al. [27], computes a flat average of held-out scores and combines it with on-task accuracy. Its held-out term is blind to what fine-tuning did to each benchmark, so a method that actively pulls damaged benchmarks back toward base and a method that simply avoids perturbing the FT distribution can post numerically similar Uni. scores even though only the first is doing the work the metric implicitly claims to measure. Our scoring conditions every held-out measurement on its FT classification ((6)): % healed is averaged only over damaged cases, % preserved only over improved ones, so *recovered-the-damage* and *never-touched-it* become two distinct statistics rather than two paths to the same number. Combined with the harmonic-mean bottleneck of §3.4, this gives the metric a guarantee an unconditional flat sum cannot match.

Per-cell aggregation. The Combined score in (7d) is for population-level (cohort) comparisons. Per-cell scores in Tab. 10 of the main paper use a separate aggregation: the harmonic mean of $(\bar{s}_{\text{method}}^{\text{held}}/\bar{s}_{\text{base}}^{\text{held}}) \times 100$ and $(s_{\text{method}}^{\text{task}}/s_{\text{FT}}^{\text{task}}) \times 100$, where $\bar{s}^{\text{held}}$ denotes the mean held-out score across the cell’s nine held-out benchmarks. Each ratio multiplies by 100 so that values cleanly exceed 100 when a method outperforms its reference (e.g., DG-Hard scores 108.5 on Qwen3.5-4B + BoolQ). Per-cell mean ratios are used here in place of the continuous % healed / % preserved statistics because, on $n \approx 9$ benchmarks per cell, the binary damaged / improved / unchanged classification produces small-sample sub-statistics that are too noisy to read cell-by-cell.

D Detailed experimental setup

This appendix documents the reproducibility-level configuration that §4 of the main paper summarizes in prose.

D.1 Fine-tuning tasks

Tab. 5 lists the seven fine-tuning tasks forming the row axis of the experimental matrix, with full HuggingFace dataset identifiers and eval split sizes. RTE, StrategyQA, ReClor, MedQA, and Winogrande use their full eval splits; BoolQ and WikiQA are subsetting to 1500 items (out of 3270 and 6165 respectively) to keep the 14×9 matrix wall-clock tractable, with per-method standard-error well below 1 pp at this sample size.

Table 5: The seven fine-tuning tasks forming the row axis of the experimental matrix. The eval split of each task becomes that cell’s on-task benchmark `task_{name}` throughout the analysis.

Task	HF dataset	Train split	Eval split	Eval n	Reasoning style
RTE	aps/super_glue (rte)	train	validation	277	binary entailment (2-class)
StrategyQA	ChilleD/StrategyQA	train	test	687	implicit multi-hop yes/no
ReClor	metaeval/reclor	train ($\sim 4.6\text{k}$)	validation	500	logical 4-way MC
BoolQ	aps/super_glue (boolq)	train (sub.)	validation	1500	reading comp. yes/no
MedQA	GBaker/MedQA-USMLE-4-options	train	test	1273	medical 4-way MC
WikiQA	wiki_qa	train	test (sub.)	1500	open-domain answer ranking
Winogrande	allenai/winogrande (x1)	train	validation	1267	pronoun-resolution 2-way

D.2 Held-out benchmarks

Tab. 6 lists the nine held-out cross-domain benchmarks evaluated identically across every (model, task, method) combination, together with their HuggingFace dataset identifiers, splits, sample sizes, shot counts, scoring metrics, and Knowledge / Cognition cohort assignment. Five benchmarks (ARC-Challenge, GSM8K, IFEval, Math-500, TruthfulQA) use their full eval splits; HellaSwag, TriviaQA,

and MNLi are subsetting as noted. MMLU uses a stratified subset of 30 questions per subject across all 57 subjects, totalling 1710 items, so each subject contributes equal weight regardless of its original size.

Table 6: The nine held-out cross-domain benchmarks evaluated identically across every (model, task, method) combination. The IFEval metric is the average of prompt $\times$ instruction strict $\times$ loose accuracy.

Benchmark	HF dataset	Eval split	Eval n	n -shot	Metric	Dimension
ARC-Challenge	allenai/ai2_arc (ARC-Challenge)	test	1172	0	accuracy	Cognition
GSM8K	gsm8k (main)	test	1319	8-shot CoT	exact-number	Cognition
HellaSwag	Rowan/hellaswag	validation (sub.)	1500	0	accuracy	Cognition
IFEval	google/IFEval	train	541	0	see caption	Cognition
Math-500	HuggingFaceH4/MATH-500	test	500	0-shot CoT	accuracy (boxed)	Cognition
MMLU	cais/mmlu (all)	test (stratified)	1710	5-shot	accuracy	Knowledge
MNLi	nyu-ml/multi_nli	validation_matched (sub.)	2000	0	accuracy	Cognition
TriviaQA	mandarjoshi/trivia_qa (rc.nocontext)	validation (sub.)	1500	0	exact-match	Knowledge
TruthfulQA	truthful_qa (multiple_choice)	validation	817	0	MCI accuracy	Knowledge

D.3 Safety benchmarks and judges

The safety axes reported in Tab. 4 of the main paper are evaluated identically across every (model, task, method) cell, with no alignment data entering the repair procedure. Generation uses the same policy as all other benchmarks: vLLM with no `max_tokens` cap (capping would bias refusal length on either side), greedy decoding, and the model’s default chat template; the prompt set and judge for each axis are summarized in Tab. 7.

Table 7: Safety benchmark configurations. HarmBench refusal is reported as the share of harmful prompts the model declines (higher is safer). XSTest over-refusal is reported on the safe half of v2 only (the “looks unsafe but isn’t” subset); lower is better. StrongREJECT reports the rubric harmfulness score on a 0-1 scale aggregated as the official convex combination of refusal, specificity, and convincingness; lower is better.

Benchmark	HF dataset	Eval n	Judge	Aggregate
HarmBench v1.0	walledai/HarmBench	200	GPT-4o classifier	refusal rate (%)
XSTest v2 (safe)	natolambert/xstest-v2-copy	250	GPT-4o 3-class [35]	over-refusal rate (%)
StrongREJECT v1.0	walledai/StrongREJECT	313	GPT-4o-mini rubric [36]	harmfulness $\in [0, 1]$

The HarmBench classifier follows the protocol of Mazeika et al. [34]: each (prompt, response) pair is classified as “refusal” or “harmful compliance” and the reported score is the refusal share. XSTest uses the safe half of v2 (250 prompts the model should comply with despite surface cues); the judge follows the three-class protocol of Röttger et al. [35] and we count any non-full-compliance (partial or refusal) as over-refusal. StrongREJECT applies its 0-1 rubric verbatim to every (prompt, response) pair; harmfulness is averaged over prompts. No category- or severity-level reweighting is applied; all benchmarks use unweighted means over their prompt sets.

D.4 Training hyperparameters

Tab. 8 lists the uniform training configuration used for every (model, task) cell, so any difference in repair outcome is attributable to the repair method rather than to training-time variance.

D.5 Repair scope

The post-hoc methods we evaluate (DG-Hard, WiSE-FT, FAPM; with TIES and DARE-Linear additionally evaluated in the App. I.3 ablation) are applied per-tensor to every weight matrix with $\text{ndim} \geq 2$ and at least 1024 elements; tensors with $\text{ndim} > 2$ are reshaped to a 2D matrix before SVD. One-dimensional parameters (biases, RMSNorm scales) are left at their fine-tuned values, since they do not admit a non-trivial Marchenko-Pastur edge. On Llama-3.2-3B-Instruct this scope covers every attention and MLP projection of the text decoder, plus `embed_tokens` and `lm_head`. On Qwen3.5-4B the same filter additionally accepts the vision-encoder weights and the multi-token-prediction head; because inference is text-only, these tensors are stripped at vLLM load

Table 8: Uniform training hyperparameters used for every (model, task) cell.

Hyperparameter	Value	Notes
Optimiser	AdamW	$\beta_1 = 0.9, \beta_2 = 0.999, \varepsilon = 10^{-8}$
Peak learning rate	1.0×10^{-5}	reached at end of warm-up
Warm-up	3% of total steps	linear from 0 to peak
LR schedule	linear decay to 0	over remaining 97% of steps
Weight decay	0	none
Per-device batch size	2	constrained by single A100-80GB
Gradient accumulation	32	effective batch = 64
Effective batch size	64	$\text{per_device} \times \text{grad_accum}$
Epochs	3	
Max sequence length	4096 tokens	task-specific within this ceiling
Precision	bf16 / fp32 Adam	
Gradient checkpointing	on for full-parameter methods	off for LoRA
Random seed	42	single seed per cell

(`checkpoint_compat.py:_DROP_PREFIXES`), and any modifications computed on them during repair never reach evaluation. The state-space `linear_attn` blocks contribute one-dimensional parameters (`A_log`, `dt_bias`) that are skipped under the dimensionality rule; their `conv1d.weight` is three-dimensional and is therefore reshaped and repaired like any other matrix. The training-time methods (L1-reg, V-SoftMask, CoFi-Tune, LoRA) see the model in their respective canonical scopes during fine-tuning and inherit no scope-restriction at inference time.

E Per-cell forgetting and balance scores

This appendix holds the two per-cell tables that the body cites in §4.2 (also labeled §4.2). Tab. 9 reports the (model, task) cell-level fine-tuning impact: how many of the nine held-out benchmarks were damaged ($\Delta_{FT} \leq -3$ pp), how many were improved ($\Delta_{FT} \geq +3$ pp), the average and worst drops over the damaged set, and the on-task gain $\Delta_{FT,task}$ (Full-SFT minus base). The remaining (9 – damaged – improved) benchmarks per cell are unchanged within ± 3 pp. Tab. 10 reports the per-cell balance score, the harmonic mean of (i) the method’s mean held-out ratio against base and (ii) the method’s on-task ratio against Full-SFT, each multiplied by 100. The per-cell view uses these mean ratios because the partition-conditional sub-statistics (% healed, % preserved) become unstable on the small per-cell $n \approx 9$. FAPM is omitted from the balance table because it never wins: its 90% entry reversion drives the held-out ratio toward base but collapses the on-task ratio, leaving it well behind both DG-Hard and WiSE-FT in every cell.

Table 9: Per-cell impact of fine-tuning across the 14 (model, task) cells of our experimental matrix. Fine-tuning damages at least one held-out benchmark in 13 of 14 cells and incidentally improves more held-out benchmarks than it damages (55 vs 30 in total).

Model	Task	# damaged ↓	# improved	Avg drop on damaged (pp)	Worst single drop (pp)	$\Delta_{FT,task}$ (pp) ↑
Qwen	RTE	2	5	−13.48	17.7 (TriviaQA)	+10.1
Qwen	StrategyQA	2	5	−4.84	6.2 (Math)	+5.2
Qwen	ReClor	3	5	−9.95	17.4 (Math)	+37.2
Qwen	BoolQ	2	5	−6.02	8.4 (Math)	+13.3
Qwen	MedQA	4	4	−33.47	91.8 (GSM8K)	+12.2
Qwen	WikiQA	2	5	−11.67	11.8 (TriviaQA)	+46.8
Qwen	Winogrande	3	3	−25.61	35.9 (GSM8K)	+25.8
Llama	RTE	3	3	−7.34	9.1 (GSM8K)	+31.0
Llama	StrategyQA	3	3	−10.36	17.0 (GSM8K)	+7.0
Llama	ReClor	0	4	(none)	(none; max −2.3 IFEval)	+19.4
Llama	BoolQ	2	2	−4.69	6.0 (IFEval)	+10.4
Llama	MedQA	1	4	−3.60	3.6 (MNLI)	+8.2
Llama	WikiQA	2	3	−5.64	7.0 (GSM8K)	+1.4
Llama	Winogrande	1	4	−7.71	7.7 (TruthfulQA)	+32.3
Total	(across 14 cells)	30	55	(mean over 30 damaged: −13.3)	(max 91.8)	(avg +18.6)

Table 10: Per-cell balance scores per (model, task) cell, plus the cohort aggregation of the winner column. The balance score is the harmonic mean of the method’s mean held-out ratio (vs base) and on-task ratio (vs FT), each multiplied by 100. FAPM [16] is excluded because it never wins on balance.

Model	Task	bal(WiSE) $\uparrow$	bal(DG) $\uparrow$	Winner
Qwen	RTE	103.6 $\pm$ 0.4	104.3$\pm$0.4	DG-Hard
Qwen	StrategyQA	102.7 $\pm$ 0.6	104.6$\pm$0.6	DG-Hard
Qwen	ReClor	98.6 $\pm$ 0.4	101.5$\pm$0.5	DG-Hard
Qwen	BoolQ	105.9 $\pm$ 0.3	108.5$\pm$0.3	DG-Hard
Qwen	MedQA	105.4 $\pm$ 0.5	105.4$\pm$0.5	tie: DG-Hard $\approx$ WiSE-FT
Qwen	WikiQA	101.9 $\pm$ 0.2	103.4$\pm$0.3	DG-Hard
Qwen	Winogrande	98.9 $\pm$ 0.3	102.8$\pm$0.3	DG-Hard
Llama	RTE	100.0$\pm$0.6	98.8 $\pm$ 0.6	WiSE-FT
Llama	StrategyQA	101.1$\pm$0.7	99.2 $\pm$ 0.7	WiSE-FT
Llama	ReClor	97.4$\pm$0.9	96.0 $\pm$ 0.8	WiSE-FT
Llama	BoolQ	100.7$\pm$0.3	100.2 $\pm$ 0.3	WiSE-FT
Llama	MedQA	102.1 $\pm$ 0.6	101.9 $\pm$ 0.6	tie: DG-Hard $\approx$ WiSE-FT
Llama	WikiQA	100.8 $\pm$ 0.3	101.3 $\pm$ 0.2	tie: DG $\approx$ WiSE-FT
Llama	Winogrande	97.4 $\pm$ 0.4	99.5$\pm$0.4	DG-Hard
<i>Aggregating the winner column:</i>				
Cohort		DG-Hard wins	WiSE-FT wins	Ties
Qwen (reasoning, $n = 7$)		6	0	1
Llama (non-reasoning, $n = 7$)		1	4	2
All cells ($n = 14$)		7	4	3

F Per-cohort Clean-up and Retention

Tabs. 11 and 12 report the per-cohort Clean-up and Retention sub-scores for every method, computed exactly as in §4.3: Clean-up = $\text{HM}(\% \text{ healed}, \text{Non-damage rate})$ and Retention = $\text{HM}(\% \text{ preserved}, \text{On-task retention})$, with the partition-conditional sub-statistics themselves clipped to $[0, 100]$ before the harmonic mean. The Knowledge and Cognition rows are the source of the per-cohort numbers cited in §4.6; the Overall row matches the population-level Tab. 1. A 0 entry in the Retention table indicates that the underlying % preserved averaged non-positive on the improved partition, so the clipped HM collapses to zero.

Table 11: Per-cohort Clean-up = $\text{HM}(\% \text{ healed}, \text{Non-damage rate})$, in $[0, 100]$, higher is better; > 100 appears when % healed overshoots base on the damaged partition. **Bold** = best per cohort; underline = second-best.

Method	Overall	Knowledge	Cognition	Reasoning	Non-reasoning
DG-Hard	83.1 $\pm$ 2.8	85.3 $\pm$ 3.2	81.6 $\pm$ 3.5	73.4 $\pm$ 2.6	92.9 $\pm$ 4.3
WiSE-FT	82.7 $\pm$ 2.7	87.5 $\pm$ 3.7	79.6 $\pm$ 3.2	80.3 $\pm$ 2.3	83.6 $\pm$ 4.8
L1-reg	83.4 $\pm$ 2.6	102.0$\pm$2.7	79.5 $\pm$ 3.1	<u>92.1$\pm$2.3</u>	75.7 $\pm$ 4.2
CoFi-Tune	77.8 $\pm$ 2.7	97.1 $\pm$ 2.1	73.7 $\pm$ 3.1	92.3$\pm$2.1	67.8 $\pm$ 4.5
V-SoftMask	24.9 $\pm$ 5.9	51.9 $\pm$ 3.4	18.3 $\pm$ 5.7	42.2 $\pm$ 3.5	0.0 $\pm$ 3.5
LoRA	<u>85.6$\pm$2.7</u>	80.5 $\pm$ 2.3	<u>86.5$\pm$3.2</u>	80.0 $\pm$ 2.5	89.4 $\pm$ 4.4
FAPM	91.9$\pm$2.7	<u>98.1$\pm$3.9</u>	90.6$\pm$3.1	87.9 $\pm$ 2.7	95.5$\pm$4.1

Per-axis trends. The two tables make the structural asymmetry of the baseline field explicit. On Retention (Tab. 12), **V-SoftMask** dominates every cohort by a wide margin, scoring ≥ 95 on all five and exceeding the next-best method by 14 to 28 pp on each. This is consistent with V-SoftMask’s mechanism: by damping gradient flow during fine-tuning it leaves the held-out and on-task distributions closer to the base, so on the improved partition it inherits the same scores it produced and thus high % preserved. The same mechanism explains its position on Clean-up (Tab. 11), where V-SoftMask is the *lowest*-scoring method on Overall (24.9) and Cognition (18.3) and falls to 0 on Non-reasoning. Because gradient damping suppresses both the noise residual and any genuine recovery, the method has nothing with which to bring damaged held-out benchmarks back

Table 12: Per-cohort Retention = $\text{HM}(\% \text{ preserved, On-task retention})$, in $[0, 100]$, higher is better. Entries of 0 indicate that the method’s average % preserved on the improved partition was non-positive, collapsing the harmonic mean to zero. **Bold** = best per cohort; underline = second-best.

Method	Overall	Knowledge	Cognition	Reasoning	Non-reasoning
DG-Hard	<u>83.5$\pm$1.6</u>	<u>84.3$\pm$1.7</u>	<u>82.5$\pm$2.4</u>	<u>89.1$\pm$1.6</u>	<u>75.3$\pm$2.5</u>
WiSE-FT	<u>71.0$\pm$2.2</u>	<u>80.3$\pm$1.9</u>	<u>57.5$\pm$3.9</u>	<u>66.4$\pm$2.7</u>	<u>76.8$\pm$2.2</u>
L1-reg	55.6 $\pm$ 3.5	78.5 $\pm$ 2.2	13.0 $\pm$ 6.1	45.0 $\pm$ 5.5	68.1 $\pm$ 3.5
CoFi-Tune	28.5 $\pm$ 6.2	58.9 $\pm$ 2.8	0.0 $\pm$ 0.2	5.2 $\pm$ 3.9	52.4 $\pm$ 4.8
V-SoftMask	97.1$\pm$1.2	97.2$\pm$1.3	97.0$\pm$1.4	95.5$\pm$1.1	99.2$\pm$1.6
LoRA	9.0 $\pm$ 3.5	40.5 $\pm$ 3.7	0.0 $\pm$ 0.0	0.0 $\pm$ 1.1	29.0 $\pm$ 6.2
FAPM	0.0 $\pm$ 0.3	0.0 $\pm$ 2.7	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 3.1

toward base, so its Clean-up component collapses. The mirror extremes are **FAPM** and **LoRA**, which post some of the highest Clean-up scores (91.9 and 85.6 Overall, respectively) yet hit 0 on every Cognition and Reasoning Retention cell because their average % preserved on the improved partition is non-positive (FAPM’s 90% sparsity reversion drags every benchmark toward base; LoRA’s adapter delta never produced the improved-partition lifts in the first place). **L1-reg** and **CoFi-Tune** are partial versions of the same single-axis pattern: high Clean-up, weak or zero Retention on Cognition.

Why balance is required. Combined = $\text{HM}(\text{Clean-up, Retention})$ is designed precisely to demote single-axis extremes via the bottleneck property of the harmonic mean: any method with a near-zero component cannot recover at the headline level regardless of how strong its other axis is. The two tables in this section make the mechanism visible cohort by cohort. V-SoftMask’s perfect Retention is undone by its near-zero Clean-up; FAPM’s strong Clean-up is undone by its identically-zero Retention; LoRA and CoFi-Tune are similarly imbalanced in less extreme but still disqualifying ways. DG-Hard is the only method that scores in the upper half of both axes simultaneously across all five cohorts, which is consistent with its leading the population-level Combined metric in Tab. 1. At the cohort level, DG-Hard is not required to win Combined everywhere (§4.6 notes that L1-reg edges it on Knowledge): the structural claim is that DG-Hard is the only method that does not collapse on either axis in any cohort, not that it wins Combined in every cohort.

G Full per-cohort sub-score breakdown

§4.4 summarized the per-cohort Combined scores for DG-Hard vs WiSE-FT. The full Clean-up and Retention sub-scores per cohort (the harmonic-mean inputs to Combined) are tabulated here. DG-Hard’s edge over WiSE-FT comes mostly from Retention: their Clean-up scores are within ~ 3 pp on most cohorts, but DG-Hard’s Retention is 12+ pp higher on Overall, Cognition, and Reasoning, driven mainly by held-out preservation since on-task retention is essentially flat at $\sim 97\%$ for both methods.

Table 13: Per-cohort Clean-up / Retention / Combined for DG-Hard vs WiSE-FT [13]. **Bold** marks the higher Combined score in this *head-to-head* DG-Hard vs WiSE-FT comparison only; it is *not* a cross-method best-per-cohort indicator. For cross-method per-cohort winners across all baselines, see App. F (Tabs. 11 and 12); per §4.6, DG-Hard wins Combined on four of five cohorts at the cross-method level (L1-reg edges it on Knowledge).

Cohort	DG Clean-up	DG Retention	DG Combined $\uparrow$	WiSE Clean-up	WiSE Retention	WiSE Combined $\uparrow$
Overall	83.1 $\pm$ 2.8	83.5 $\pm$ 1.6	83.3	82.7 $\pm$ 2.7	71.0 $\pm$ 2.2	76.4
Knowledge	85.3 $\pm$ 3.2	84.3 $\pm$ 1.7	84.8	87.5 $\pm$ 3.7	80.3 $\pm$ 1.9	83.7
Cognition	81.6 $\pm$ 3.5	82.5 $\pm$ 2.4	82.1	79.6 $\pm$ 3.2	57.5 $\pm$ 3.9	66.8
Reasoning (Qwen)	73.4 $\pm$ 2.6	89.1 $\pm$ 1.6	80.5	80.3 $\pm$ 2.3	66.4 $\pm$ 2.7	72.7
Non-reasoning (Llama)	92.9 $\pm$ 4.3	75.3 $\pm$ 2.5	83.2	83.6 $\pm$ 4.8	76.8 $\pm$ 2.2	80.1

H Per-(model, task, method, benchmark) results

For full transparency, Tab. 14 and Tab. 15 list every per-cell, per-benchmark score in the experimental matrix (1134 measurements in total: 2 models $\times$ 7 tasks $\times$ 9 method rows $\times$ 9 held-out benchmarks;

each table additionally reports 3 derived columns (*Avg.*, *Results*, *Combined*) per row). Each task block contains nine method rows (Pre-trained, Full-SFT, six repair baselines, plus our DG-Hard); within each task block, **bold** marks the best score per column and underline marks the second-best (when separated by ≥ 0.001). The right-most three columns are *Avg.* (mean of the nine held-out benchmarks), *Results* (on-task task_{name} score), and *Combined* (defined in the table caption).

H.1 Method-specific configurations

L1-reg adds an L1 penalty $\|W - W_{\text{base}}\|_1$ to the loss with $\lambda = 1.0 \times 10^{-6}$. **LoRA** uses rank $r = 16$, $\alpha = 32$, dropout 0.05, and auto-selected target modules. **WiSE-FT** sets the mixing weight $\alpha = 0.5$, applied post-hoc on the Full-SFT checkpoint to give the linear weight average $W^* = \frac{1}{2}W_{\text{base}} + \frac{1}{2}W_{\text{ft}}$, the canonical setting from Wortsman et al. [13]. **FAPM** reverts the entries of Δ ranked lowest under the forgetting-aware FAPM score, which combines absolute change with a relative-change penalty against W_{base} ; we use the published 90% reversion rate. **V-SoftMask** (our reproduction of DAS, 23) uses a calibration batch size of 4 and applies mask $= (1 - \text{normalised gradient importance})$, fixed for the SFT run. **CoFi-Tune** uses the two-stage configuration of Zhang et al. [27] verbatim (full mechanism in §2); we run the coarse layer-range filter at the second-quartile setting ($N \times 25\%$, $N \times 50\%$). **DG-Hard** applies the $\omega(\beta) \cdot \hat{\sigma}_{\text{eff}}$ hard threshold of §3; it has no tuning knobs and uses no calibration data. **TIES** [15] keeps the top- k entries of $|\Delta|$ by magnitude, sign-elects, and averages; we use `keep_ratio` = 0.2 ($k = 20\%$) with mixing weight $\lambda = 1$. **DARE** [14] drops entries of Δ with probability p and rescales the survivors by $1/(1 - p)$; we use `drop_prob` = 0.5 with the rescaled (rather than raw) variant. TIES and DARE are evaluated only in the App. I.3 ablation, not in the headline comparison of Tab. 14 and Tab. 15.

H.2 Inference engine and decoding

All inference uses vLLM with `-enforce-eager`, `max_model_len=8192`, `dtype=bfloat16`, greedy decoding (temperature = 0), *no* `max_tokens` cap, and the model’s native chat template applied via the tokenizer’s `apply_chat_template` method. Greedy decoding under vLLM’s continuous batching is not bit-deterministic: the same prompt can produce slightly different completions on re-runs, with a measured **drift floor of about ± 1 pp per benchmark**. We treat any delta smaller than this floor as noise, and use a **3 pp significance threshold** throughout the paper. The threshold is conservative ($\approx 3 \times$ the drift floor) and is chosen to capture genuine but moderate fine-tuning effects while excluding sampling drift. Output scoring strips `<think>... </think>` blocks and reasoning preludes before metric computation.

I Other experiments and ablations

This appendix collects three experiments supporting the design choices of DG-Hard. App. I.1 quantifies which weight classes carry the noise mass that DG-Hard removes (layer-class noise concentration); App. I.2 verifies the layer-mask choice via a causal test; App. I.3 compares against element-wise merging baselines at canonical and matched-rollback operating points.

I.1 Layer-wise noise concentration

For each (model, task) fine-tune, we cache the SVD of every 2D weight delta and classify each matrix by its layer class (e.g., `mlp.up_proj`, `attn.q_proj`). Per class we report the mean noise-energy fraction (the share of singular-value-squared mass sitting below the MP edge λ_{MP} of (3)) and the total norm-share (the fraction of the network’s Frobenius² delta mass attributable to the class). Their product is the operationally meaningful noise mass per class under the signal-plus-noise model of (2). Averaged across six (model, task) pairs, the MLP gate and up projections concentrate the bulk of the noise mass that DG-Hard removes (Tab. 16); attention and embedding classes contribute substantially less.

Metric note. The mean-noise column above (energy fraction of squared singular values below λ_{MP}) and the MP-fit score of (8) (a Kolmogorov-Smirnov distance between the empirical and theoretical CDFs) measure different things: the former is the share of squared mass under the bulk edge, the latter is how closely the spectrum’s shape matches the MP density. The numerical proximity of

Table 14: Per-benchmark held-out scores for Qwen3.5-4B across all (task, method) cells. **Bold** = best per column within the task block; underline = second-best (gap ≥ 0.001 from best). Avg. = mean of the nine held-out benchmarks. Results = on-task task_{name} score. Combined = HM(% healed, % preserved) on the partitioned held-out set, with per-benchmark % healed and % preserved clipped to $[0, 100]$ before averaging; cells with no damaged or no improved benchmarks default the corresponding side to 100. Pre-trained and Full-SFT have no defined Combined (they parameterize the partition). Bold/underline on Combined follow the same convention as other columns, restricted to the seven repair methods (excluding the two reference rows). “Repair methods” here covers both training-time interventions (L1-reg, V-SoftMask [23], CoFi-Tune [27], LoRA [18]) and post-hoc methods (WiSE-FT [13], FAPM [16], DG-Hard); see §2 for the full bucketing.

Task	Method	ARC	GSM8K	HSwag	IFEval	Math	MMLU	MNLI	Trivia	TruthQA	Avg.	Results	Combined
MedQA	Pre-trained	0.8643	0.9295	0.5060	0.3460	0.5320	0.6041	0.8464	0.4767	0.6548	0.6400	0.6308	—
	Full-SFT	0.9138	0.0114	0.8780	0.3026	0.2640	0.7550	0.8270	0.3673	0.7552	0.5638	0.7526	—
	L1-reg	0.6374	0.9401	0.4433	0.3504	<u>0.5920</u>	0.6526	<u>0.8425</u>	0.5020	0.6304	0.6212	0.7494	0.1489
	WiSE-FT	0.8993	0.8590	<u>0.8787</u>	0.3551	0.5920	0.7444	0.8185	0.4967	<u>0.7625</u>	0.7118	0.7541	0.9437
	V-SoftMask	<u>0.9078</u>	0.1099	0.8760	0.3140	0.2520	<u>0.7526</u>	0.8265	0.3887	0.7491	0.5752	0.7423	0.2459
	CoFi-Tune	0.7799	<u>0.9348</u>	0.4667	0.3554	0.5500	0.6240	0.8365	0.4833	0.6622	0.6325	0.7494	0.0975
	LoRA	0.7526	0.9022	0.5413	0.3514	0.5160	0.6240	0.8115	0.4600	0.6255	0.6205	0.7321	0.1069
	FAPM	0.8686	0.9280	0.5013	0.3458	0.4980	0.6047	0.8370	0.4673	0.6438	0.6327	0.6088	0.0440
	DG-Hard (Ours)	0.9019	0.8158	0.8820	0.3501	0.6000	0.7462	0.8120	0.5027	0.7797	<u>0.7100</u>	0.7549	0.9466
WikiQA	Pre-trained	0.8643	0.9295	0.5060	0.3460	0.5320	0.6041	0.8464	0.4767	0.6548	0.6400	0.4973	—
	Full-SFT	0.8985	<u>0.9424</u>	<u>0.8527</u>	0.6053	0.5340	0.7971	0.7310	0.3587	<u>0.6965</u>	<u>0.7129</u>	0.9651	—
	L1-reg	0.8353	0.9280	0.4987	0.3496	0.5020	0.6965	0.8250	0.4867	0.7173	0.6488	0.9674	0.4492
	WiSE-FT	0.8072	0.9447	0.5880	0.3670	<u>0.5520</u>	0.6930	0.8225	0.5293	0.6891	0.6659	0.9630	<u>0.4720</u>
	V-SoftMask	<u>0.8805</u>	0.9393	0.8620	<u>0.5772</u>	0.5580	0.7971	0.7550	0.4087	0.6756	0.7171	0.9659	0.4486
	CoFi-Tune	0.8524	0.9386	0.5460	0.3526	0.5060	0.6684	0.8180	0.5040	0.6573	0.6493	0.9648	0.1900
	LoRA	0.8652	0.9318	0.5167	0.3478	0.5240	0.6433	0.7905	0.4707	0.6769	0.6407	0.9679	0.2613
	FAPM	0.8652	0.9325	0.4820	0.3524	0.4860	0.6123	<u>0.8420</u>	0.4693	0.6389	0.6312	0.4986	0.0360
	DG-Hard (Ours)	0.7867	0.9424	0.7953	0.4555	0.5380	0.7316	0.7650	0.4893	0.6928	0.6885	0.9613	0.6039
Winogrande	Pre-trained	<u>0.8643</u>	<u>0.9295</u>	0.5060	0.3460	0.5320	0.6041	<u>0.8464</u>	0.4767	0.6548	0.6400	0.6393	—
	Full-SFT	0.6305	0.5709	<u>0.8727</u>	<u>0.4166</u>	0.5520	0.7520	0.8170	0.3007	0.6573	0.6189	<u>0.8974</u>	—
	L1-reg	0.8302	0.9386	0.4407	0.3511	0.5700	0.6690	0.8530	<u>0.4973</u>	<u>0.6842</u>	0.6482	0.8627	0.2888
	WiSE-FT	0.8063	0.9174	0.6027	0.3619	<u>0.5600</u>	0.7058	0.8305	0.4900	0.6267	0.6557	0.8579	0.5473
	V-SoftMask	0.8336	0.8287	0.8753	0.5039	0.5600	0.7930	0.8250	0.4353	0.6450	0.7000	0.8998	0.8790
	CoFi-Tune	0.8183	0.9212	0.4620	0.3492	0.5200	0.6567	0.8435	0.4727	0.7001	0.6382	0.8477	0.2331
	LoRA	0.8038	0.9052	0.0613	0.3510	0.5120	0.6713	0.7200	0.4513	0.4945	0.5523	0.8548	0.2901
	FAPM	0.8763	0.9265	0.4807	0.3481	0.5340	0.6246	0.8455	0.4813	0.6463	0.6404	0.6385	0.1059
	DG-Hard (Ours)	0.7415	0.8795	<u>0.8527</u>	0.3656	0.5500	<u>0.7661</u>	0.8405	0.5053	0.6095	<u>0.6790</u>	0.8950	<u>0.7592</u>
StrategyQA	Pre-trained	0.8643	0.9295	0.5060	0.3460	0.5320	0.6041	0.8464	0.4767	0.6548	0.6400	0.7176	—
	Full-SFT	0.8959	0.8946	0.8713	<u>0.6361</u>	0.4700	0.7661	0.8278	0.4740	<u>0.7026</u>	0.7265	0.7700	—
	L1-reg	0.8874	0.9128	0.8500	0.5273	0.4840	0.7620	0.8279	0.4560	0.6830	0.7100	<u>0.7584</u>	0.5037
	WiSE-FT	0.8951	0.9265	0.8353	0.3952	0.5180	0.7526	0.8323	0.4453	0.6659	0.6962	0.7496	0.7267
	V-SoftMask	0.8959	0.8939	<u>0.8653</u>	0.6553	0.4560	0.7661	0.8258	0.4627	0.6891	<u>0.7233</u>	0.7496	0.0000
	CoFi-Tune	0.8003	0.9393	0.5267	0.3691	0.4840	0.6199	0.8531	<u>0.4887</u>	0.7087	0.6433	0.7555	0.3518
	LoRA	0.8635	0.9287	0.5027	0.3470	0.5460	0.6205	<u>0.8486</u>	0.4933	0.6622	0.6458	0.6652	0.0982
	FAPM	0.8823	0.9265	0.4873	0.2106	0.3800	0.6287	0.8454	0.4600	0.6512	0.6080	0.7132	0.2188
	DG-Hard (Ours)	0.8942	0.9105	0.8547	0.6003	0.4900	0.7632	0.8340	0.4600	0.6928	0.7222	0.7511	<u>0.5457</u>
BoolQ	Pre-trained	0.8643	0.9295	0.5060	0.3460	0.5320	0.6041	0.8464	0.4767	0.6548	0.6400	0.7847	—
	Full-SFT	0.9070	0.9121	0.8807	0.7440	0.4480	0.7731	0.8099	0.4760	0.7809	0.7480	0.9177	—
	L1-reg	0.7722	0.9477	0.5087	0.3722	<u>0.5780</u>	0.7275	0.8399	0.5320	0.7748	0.6726	0.9180	0.5068
	WiSE-FT	0.7944	<u>0.9401</u>	0.8653	0.4914	0.5500	0.7620	0.8279	<u>0.5080</u>	0.7405	0.7200	0.9180	0.6579
	V-SoftMask	0.9121	0.9121	0.8853	<u>0.7194</u>	0.4620	<u>0.7754</u>	0.8066	0.4747	0.7846	<u>0.7480</u>	0.9165	0.1537
	CoFi-Tune	0.7910	0.9249	0.5180	0.3618	0.5120	0.6585	0.8420	0.5067	0.7234	0.6487	0.9116	0.3052
	LoRA	0.8635	0.9378	0.5033	0.3412	0.5420	0.6327	0.8435	0.5073	0.6805	0.6502	0.9080	0.1386
	FAPM	0.8771	0.9303	0.4913	0.3514	0.5260	0.6094	0.8465	0.4640	0.6671	0.6403	0.7881	0.1618
	DG-Hard (Ours)	0.9053	0.9249	0.8793	0.6475	0.6180	0.8053	0.8070	0.4640	0.7687	0.7578	0.9187	<u>0.6487</u>
ReClor	Pre-trained	0.8643	0.9295	0.5060	0.3460	0.5320	0.6041	0.8464	0.4767	0.6548	0.6400	0.5120	—
	Full-SFT	0.9121	0.8529	0.8740	0.4442	0.3580	<u>0.7626</u>	0.7986	0.4593	0.7234	0.6872	<u>0.8840</u>	—
	L1-reg	0.9087	0.9234	0.8773	0.3682	0.4080	0.7620	0.8193	<u>0.4893</u>	<u>0.7442</u>	0.7000	0.8660	0.6595
	WiSE-FT	0.9087	0.7506	0.8780	0.3086	0.3920	0.7556	0.8164	0.2587	0.7185	0.6430	0.8560	0.3036
	V-SoftMask	0.9053	0.8696	0.8707	<u>0.4445</u>	0.3420	0.7661	0.7955	0.4733	0.7271	<u>0.6882</u>	0.8900	0.1351
	CoFi-Tune	0.9061	0.9363	0.8240	0.2495	0.3860	0.6930	0.7948	0.5047	0.7601	0.6727	0.8200	0.4879
	LoRA	0.8251	0.9037	0.8220	0.2193	0.3780	0.6807	0.8007	0.4780	0.7393	0.6497	0.8080	0.3460
	FAPM	0.8677	0.9363	0.4993	0.3553	0.5460	0.6018	<u>0.8385</u>	0.4707	0.6499	0.6406	0.4860	0.0640
	DG-Hard (Ours)	<u>0.9096</u>	0.8249	0.8767	0.4543	0.5180	0.7591	0.8085	0.2887	0.7283	0.6853	0.8520	<u>0.5440</u>
RTE	Pre-trained	<u>0.8643</u>	0.9295	0.5060	0.3460	0.5320	0.6041	0.8464	0.4767	0.6548	0.6400	0.8484	—
	Full-SFT	0.7713	0.9242	0.8847	<u>0.4150</u>	0.6140	0.7901	0.8202	0.3000	0.7356	0.6950	0.9495	—
	L1-reg	0.7841	0.9348	0.8773	0.4327	<u>0.5980</u>	0.7538	0.8508	0.4780	0.7173	0.7141	0.9386	0.6887
	WiSE-FT	0.8157	<u>0.9409</u>	0.7773	0.3883	0.5640	0.7082	0.8455	<u>0.4927</u>	0.7356	0.6965	0.9386	0.6948
	V-SoftMask	0.8635	0.9348	0.7847	0.3547	0.5340	0.6766	0.8500	0.3767	0.7491	0.6804	0.9314	0.5554
	CoFi-Tune	0.8456	0.9439	0.5187	0.3488	0.4780	0.6889	0.8415	0.5153	0.7136	0.6549	0.9386	0.3929
	LoRA	0.8686	0.9318	0.4927	0.3529	0.5180	0.6088	0.8413	0.4787	0.6499	0.6381	0.9242	0.0486
	FAPM	0.8609	0.9393	0.4953	0.2288	0.3640	0.6251	0.8469	0.4760	0.6646	0.6112	0.8484	0.0895
	DG-Hard (Ours)	0.7517	0.9401	<u>0.8793</u>	0.4113	0.5900	<u>0.7766</u>	0.8425	0.4760	<u>0.7417</u>	<u>0.7121</u>	0.9314	0.6447

Table 15: Per-benchmark held-out scores for Llama-3.2-3B-Instruct across all (task, method) cells. **Bold** = best per column within the task block; underline = second-best (gap ≥ 0.001 from best). Avg. = mean of the nine held-out benchmarks. Results = on-task task_{name} score. Combined = HM(% healed, % preserved) on the partitioned held-out set, with per-benchmark % healed and % preserved clipped to [0, 100] before averaging. Pre-trained and Full-SFT have no defined Combined. Bold/underline on Combined follow the same convention as other columns, restricted to the seven repair methods (excluding the two reference rows). “Repair methods” here covers both training-time interventions (L1-reg, V-SoftMask [23], CoFi-Tune [27], LoRA [18]) and post-hoc methods (WiSE-FT [13], FAPM [16], DG-Hard); see §2 for the full bucketing.

Task	Method	ARC	GSM8K	HSwag	IFEval	Math	MMLU	MNLI	Trivia	TruthQA	Avg.	Results	Combined
MedQA	Pre-trained	<u>0.7543</u>	0.7218	0.6193	0.7784	0.4200	0.3474	<u>0.5025</u>	0.3360	0.5606	0.5600	0.4949	—
	Full-SFT	0.7440	0.7058	<u>0.6980</u>	0.7599	0.4180	<u>0.5655</u>	0.4665	0.5253	<u>0.6157</u>	<u>0.6110</u>	<u>0.5774</u>	—
	L1-reg	0.7457	0.7225	0.6787	0.7722	0.3980	0.4094	0.4745	0.4367	0.5973	0.5817	0.5695	<u>0.3181</u>
	WiSE-FT	0.7517	0.7202	0.6727	0.7667	0.4120	0.3860	0.4710	0.4267	0.5961	0.5781	0.5829	0.1996
	V-SoftMask	0.7474	0.7096	0.7007	0.7557	<u>0.4360</u>	0.5719	0.4610	0.5247	0.6108	0.6131	0.5742	0.0000
	CoFi-Tune	0.7534	0.7134	0.6460	0.7740	0.4620	0.3585	0.4415	0.3900	0.6059	0.5716	0.5687	0.0000
	LoRA	0.7483	0.7339	0.6440	0.7612	0.4300	0.3953	0.4740	0.4027	0.6193	0.5788	0.5734	0.2890
	FAPM	0.7594	0.7218	0.6180	<u>0.7766</u>	0.4220	0.3538	0.5005	0.3427	0.5545	0.5610	0.5035	0.0318
	DG-Hard (Ours)	0.7449	<u>0.7301</u>	0.6693	0.7666	0.4280	0.3848	0.5045	0.4113	0.5924	0.5813	0.5774	0.6166
WikiQA	Pre-trained	<u>0.7543</u>	0.7218	0.6193	<u>0.7784</u>	0.4200	0.3474	0.5025	0.3360	0.5606	0.5600	0.9526	—
	Full-SFT	0.7474	0.6520	0.5947	0.7354	0.4320	0.3404	<u>0.5580</u>	0.4933	0.6108	<u>0.5738</u>	0.9664	—
	L1-reg	0.7389	0.7210	0.5347	0.7703	0.4200	0.3404	0.4920	0.4320	0.5655	0.5572	0.9638	0.3738
	WiSE-FT	0.7423	0.6808	0.5947	0.7535	0.4340	<u>0.3491</u>	0.5425	0.4547	0.5936	0.5717	0.9629	0.5251
	V-SoftMask	0.7381	0.6368	0.5920	0.7423	0.4380	0.3392	0.5600	0.4900	0.6083	0.5716	0.9663	0.1475
	CoFi-Tune	0.7193	0.6854	0.5687	0.7825	0.4200	0.3058	0.4790	0.3813	<u>0.6095</u>	0.5502	0.9658	<u>0.5366</u>
	LoRA	0.7329	0.7180	0.5687	0.7644	<u>0.4360</u>	0.3450	0.4705	0.4380	0.5802	0.5615	0.9646	0.4850
	FAPM	0.7688	<u>0.7225</u>	<u>0.6127</u>	0.7743	0.4320	0.3532	0.5050	0.3407	0.5557	0.5628	0.9531	0.0485
	DG-Hard (Ours)	0.7415	0.7293	0.5833	0.7720	0.4220	0.3433	0.5535	0.4833	0.5961	0.5805	0.9580	0.8881
Winogrande	Pre-trained	0.7543	0.7218	0.6193	0.7784	0.4200	0.3474	0.5025	0.3360	<u>0.5606</u>	0.5600	0.5209	—
	Full-SFT	<u>0.7765</u>	0.6960	0.6673	0.7647	0.4140	<u>0.4225</u>	0.5370	0.4533	0.4835	0.5800	0.8437	—
	L1-reg	0.7167	0.7134	0.5947	0.7524	0.4120	0.3222	0.5150	0.4487	0.5422	0.5575	0.7956	0.4612
	WiSE-FT	0.7671	0.6967	0.6547	0.7715	0.4320	0.3731	0.5375	<u>0.4627</u>	0.5483	<u>0.5826</u>	0.7719	<u>0.8010</u>
	V-SoftMask	0.7782	0.7058	0.6673	0.7572	0.4100	0.4351	0.5365	0.4507	0.4786	0.5799	<u>0.8421</u>	0.0000
	CoFi-Tune	0.7218	0.6960	0.6193	0.7737	0.4320	0.3427	0.5765	0.3920	0.5692	0.5692	0.7301	0.5394
	LoRA	0.7560	0.6922	0.6173	0.7874	0.4320	0.3719	0.4900	0.3707	0.4933	0.5567	0.8114	0.1377
	FAPM	0.7585	<u>0.7233</u>	0.6160	<u>0.7800</u>	0.4240	0.3515	0.5035	0.3393	0.5557	0.5613	0.5178	0.0527
	DG-Hard (Ours)	0.7517	0.7339	0.6580	0.7713	0.4060	0.3854	<u>0.5475</u>	0.4680	0.5447	0.5852	0.8011	0.8066
StrategyQA	Pre-trained	0.7543	<u>0.7218</u>	<u>0.6193</u>	0.7784	0.4200	0.3474	0.5025	0.3360	0.5606	0.5600	0.6521	—
	Full-SFT	0.7457	0.5519	0.6220	0.6854	0.3720	0.5228	0.5235	0.5427	0.5936	0.5733	<u>0.7220</u>	—
	L1-reg	0.7398	0.6740	0.6040	0.7228	0.4000	0.4766	0.5130	0.5213	0.6010	0.5836	0.7234	0.6898
	WiSE-FT	0.7363	0.6998	0.6100	0.7334	0.4380	0.4211	0.5190	0.4940	0.6083	0.5844	0.7074	0.7604
	V-SoftMask	0.7526	0.5557	0.6173	0.6808	0.3960	<u>0.5187</u>	0.5320	<u>0.5393</u>	0.5924	0.5761	0.7176	0.2954
	CoFi-Tune	0.7227	0.6884	0.5753	0.7708	0.4200	0.3491	0.5165	0.4427	<u>0.6059</u>	0.5657	0.6929	0.6519
	LoRA	0.7389	0.7240	0.6020	<u>0.7730</u>	0.4400	0.3164	0.5070	0.3687	0.5594	0.5588	0.6623	0.1000
	FAPM	0.7551	0.7187	0.6153	0.7729	0.4160	0.3485	0.5050	0.3367	0.5508	0.5577	0.6463	0.0066
	DG-Hard (Ours)	0.7270	0.7058	0.6080	0.7324	0.4400	0.3772	<u>0.5250</u>	0.4767	0.5973	0.5766	0.6914	<u>0.6980</u>
BoolQ	Pre-trained	0.7543	0.7218	<u>0.6193</u>	0.7784	0.4200	0.3474	<u>0.5025</u>	0.3360	0.5606	0.5600	0.7859	—
	Full-SFT	0.7628	0.6876	0.6353	0.7187	0.4040	0.4801	0.4900	0.4967	0.5679	0.5826	0.8902	—
	L1-reg	0.7543	0.7233	0.5520	<u>0.7756</u>	0.4160	0.3661	0.4845	0.4807	<u>0.5887</u>	0.5712	0.8872	0.6793
	WiSE-FT	<u>0.7551</u>	0.7240	0.6120	0.7690	0.4240	0.3813	0.4995	0.4647	0.5875	0.5797	0.8731	<u>0.6714</u>
	V-SoftMask	0.7517	0.6657	0.6067	0.7046	0.4440	<u>0.4520</u>	0.4715	<u>0.4847</u>	0.5875	0.5743	0.8951	0.0000
	CoFi-Tune	0.7346	0.7119	0.5687	0.7681	0.4340	0.3105	0.4555	0.4313	0.5985	0.5570	0.8832	0.4282
	LoRA	0.7423	0.7377	0.5607	0.7606	<u>0.4400</u>	0.3298	0.4685	0.4567	0.5789	0.5639	0.8743	0.5211
	FAPM	0.7551	0.7149	0.6167	0.7693	0.4140	0.3538	0.5000	0.3387	0.5532	0.5573	0.7856	0.0626
	DG-Hard (Ours)	0.7526	<u>0.7286</u>	0.6147	0.7686	0.4280	0.3673	0.5065	0.4720	0.5802	<u>0.5798</u>	0.8633	0.6457
ReClor	Pre-trained	0.7543	0.7218	0.6193	<u>0.7784</u>	0.4200	0.3474	0.5025	0.3360	0.5606	0.5600	0.4860	—
	Full-SFT	<u>0.7628</u>	<u>0.7468</u>	0.6573	0.7558	0.4300	0.4386	<u>0.5290</u>	0.4860	0.6255	0.6035	0.6800	—
	L1-reg	0.7526	0.7354	0.6420	0.7548	0.4460	0.3731	0.5330	0.4427	0.6181	0.5886	0.6760	0.7648
	WiSE-FT	0.7688	0.7263	0.6373	0.7856	0.4500	0.3532	0.5155	0.4467	0.6083	0.5880	0.6180	0.6692
	V-SoftMask	0.7628	0.7521	0.6573	0.7492	0.4320	<u>0.4310</u>	0.5260	<u>0.4780</u>	<u>0.6230</u>	<u>0.6013</u>	0.6800	0.9777
	CoFi-Tune	0.7355	0.7233	0.6053	0.7689	0.4360	0.3310	0.5260	0.3907	0.6157	0.5703	0.5960	0.4655
	LoRA	0.7415	0.7233	0.6140	0.7764	0.4420	0.3257	0.5155	0.3640	0.5887	0.5657	0.5620	0.2686
	FAPM	0.7560	0.7187	0.6187	0.7740	0.4240	0.3596	0.5030	0.3453	0.5508	0.5611	0.4900	0.0938
	DG-Hard (Ours)	0.7560	0.7324	0.6340	0.7733	0.4420	0.3386	0.5255	0.4567	0.6059	0.5849	0.6040	0.6414
RTE	Pre-trained	<u>0.7543</u>	<u>0.7218</u>	0.6193	<u>0.7784</u>	0.4200	0.3474	0.5025	0.3360	0.5606	0.5600	0.5957	—
	Full-SFT	0.7346	0.6308	0.6013	0.7031	0.3660	0.5228	0.5870	0.5053	0.5802	0.5812	0.9061	—
	L1-reg	0.7082	0.6960	0.5847	0.7508	0.3980	0.4398	<u>0.5895</u>	0.4693	0.6034	<u>0.5822</u>	0.8845	0.7042
	WiSE-FT	0.7142	0.7051	0.5853	0.7686	0.4180	0.3889	0.5630	0.4620	0.6010	0.5784	0.8773	0.6895
	V-SoftMask	0.7321	0.6240	0.6087	0.7003	0.3800	<u>0.5187</u>	0.6050	0.5053	0.5924	0.5852	0.9061	0.1590
	CoFi-Tune	0.7014	0.6808	0.5600	0.7534	<u>0.4280</u>	0.3205	0.5225	0.4160	0.6010	0.5537	0.8736	0.3582
	LoRA	0.7321	0.7309	0.6053	0.7860	0.4220	0.3392	0.5320	0.3727	0.5753	0.5662	0.8448	0.3173
	FAPM	0.7568	0.7165	<u>0.6127</u>	0.7750	0.4080	0.3520	0.5045	0.3407	0.5618	0.5587	0.5993	0.0505
	DG-Hard (Ours)	0.7201	0.7081	0.5840	0.7726	0.4300	0.3667	0.5620	0.4467	0.6034	0.5771	0.8592	0.6398

Table 16: Per-class noise ranking, averaged over six (model, task) pairs. Mean noise is the fraction of squared singular-value mass below λ_{MP} . Mean norm-share is the class share of the network’s total Frobenius² delta. Their product is the noise mass per class. Top-two rows in bold.

Class	Mean noise	Mean norm-share	Noise $\times$ norm-share
mlp.up_proj	0.93	0.231	0.213
mlp.gate_proj	0.93	0.214	0.196
mlp.down_proj	0.55	0.238	0.131
attn.v_proj	0.41	0.045	0.018
attn.o_proj	0.10	0.099	0.010
attn.k_proj	0.37	0.022	0.008
embed_tokens	≈ 1.00	0.026	0.026

the mlp.up_proj row’s 0.93 to the 93.1% MP-fit reported in App. A is therefore coincidental, not the same quantity. The embed_tokens value ≈ 1.00 is a degenerate case in which essentially all of the FT delta’s spectral mass sits under the bulk edge: there is no task-specific signal in the embedding update on these (model, task) pairs, consistent with App. A’s claim that the learned signal is concentrated in a small number of spike directions, just not in the embedding for these fine-tunes.

The aggregate ranking conceals a sharp cross-family asymmetry in attention noise (Tab. 17): on Llama-3.2-3B, attn.q_proj has mean below-edge mass 0.025, while on Qwen3.5-4B the same class shows 0.91, a $36\times$ ratio. The attn.o_proj class shows a $7\times$ ratio in the same direction. MLP classes are family-invariant (ratios within 1.16).

Table 17: Cross-family asymmetry in mean noise-energy fraction. Attention classes diverge sharply between Llama-3.2-3B-Instruct and Qwen3.5-4B; MLP classes do not.

Class	Llama mean noise	Qwen mean noise	Ratio
mlp.gate_proj	0.93	0.94	1.01
mlp.up_proj	0.92	0.93	1.01
mlp.down_proj	0.51	0.59	1.16
attn.q_proj	0.025	0.91	$36\times$
attn.o_proj	0.027	0.19	$7\times$
attn.k_proj	0.36	0.39	1.08
attn.v_proj	0.44	0.35	0.80

The family asymmetry is consistent with the hypothesis that reasoning post-training (long chain-of-thought RL or distillation, present in Qwen3.5-4B but not in Llama-3.2-3B-Instruct) disperses attention weights and leaves more below-edge mass in the FT delta, but this explanation is preliminary and untested in the present run. Regardless of the underlying cause, the gate and up MLP classes carry the dominant share of removable noise mass in both families, motivating the gate+up layer mask whose causal effect is tested in §I.2.

I.2 Layer-mask causal test

App. I.1 ranks 2D weight classes by noise mass and identifies gate_proj and up_proj as the top-two carriers on every (model, task) pair. This appendix verifies that ranking causally: we vary the set of layers pruned by DG-Hard and measure cross-domain accuracy. The protocol is DG-Hard canonical (scale = 1.0, DG σ) on two pairs, Qwen3.5-4B with Winogrande (severe FT damage) and Llama-3.2-3B with MedQA (mild FT damage), under four masks: ALL (every 2D matrix), mlp_only ($\backslash .\text{mlp}\backslash .$), attn_only ($\backslash .\text{self_attn}\backslash .$), and gate_up ($\backslash .\text{mlp}\backslash .(\text{gate|up})_proj$). Cells outside the active mask pass through with the full FT delta (no shrinkage); we report Frobenius retention r alongside benchmark accuracies and the protocol-aligned Combined score. The four masks above target attention/MLP partitions of the decoder; the embed_tokens and lm_head matrices, which are in scope for the headline DG-Hard configuration (App. D.5), are kept at their ALL-mask repaired state in all four conditions of this ablation. The focus of this experiment is the attention-vs-MLP partition; the embedding contribution is characterised separately in App. I.1.

Column conventions. `task_win` and `task_med` are on-task accuracy on Winogrande and MedQA, respectively. The Combined column applies the partition-conditional harmonic mean of (7d); when $U = \emptyset$ (Tab. 18) the non-damage term of (7) is undefined (0/0) and we adopt the convention non-damage = 100, matching the implementation. With this convention the `attn_only` row of Tab. 18 collapses to Combined = 0.0 because percent-healed is negative on the damaged benches (GSM8K, ARC-C), which the harmonic mean drives to zero independently of the U-term.

Table 18: Mask ablation on Qwen3.5-4B + Winogrande (severe FT damage). $D = \{\text{arc_challenge, gsm8k}\}$, $I = \{\text{hellaswag, ifeval, mmlu}\}$, $U = \emptyset$. Combined uses the protocol harmonic mean over the partition. Winning mask on Combined in bold.

Mask	GSM8K	ARC-C	HSwag	IFEval	MMLU	task_win	r	Combined
ALL	87.7	73.9	85.0	36.1	75.7	89.4	0.314	72.6
mlp_only	89.2	72.6	86.7	35.7	78.2	89.0	0.633	73.7
attn_only	58.5	55.3	87.2	41.2	75.0	89.7	0.966	0.0
gate_up	87.6	67.7	87.0	39.4	77.9	89.6	0.780	67.4

Table 19: Mask ablation on Llama-3.2-3B + MedQA (mild FT damage). $D = \{\text{gsm8k}\}$, $I = \{\text{hellaswag, mmlu}\}$, $U = \{\text{arc_challenge, ifeval}\}$. Winning mask on Combined in bold.

Mask	GSM8K	ARC-C	HSwag	IFEval	MMLU	task_med	r	Combined
ALL	73.8	74.6	67.1	75.9	38.7	57.7	0.186	79.5
mlp_only	71.0	74.3	68.1	75.4	43.0	57.3	0.540	79.5
attn_only	70.4	73.9	69.6	75.9	49.2	57.7	0.874	82.9
gate_up	69.7	74.4	69.5	76.9	49.1	57.5	0.730	75.3

Pattern A (Qwen, severe FT). `ALL`, `mlp_only`, and `gate_up` cluster closely on the per-benchmark damaged accuracies (GSM8K 87.6–89.2, ARC-C 67.7–73.9); `attn_only` collapses on the damaged benches with GSM8K dropping to 58.5 (vs. 87.7 under `ALL`) and ARC-C to 55.3 (vs. 73.9). The protocol-aligned Combined column is more discriminating still: `attn_only` scores 0.0 because it has negative percent healed on the damaged benches, which the harmonic mean penalises. Repairing only attention while leaving the MLP delta unshrunk keeps too much noise in the model, regardless of how aggressively attention is repaired. Repairing the MLP class is necessary; repairing only attention is insufficient.

Pattern B (Llama, mild FT). All four masks cluster tightly across the per-benchmark accuracies, and Combined separates them only mildly. With a mild FT delta, the absolute amount of noise to remove is small, so any reasonable repair yields similar cross-domain accuracy. We treat this as a *negative* result: the variance is genuinely small, not evidence that all masks are equal. The cross-family contrast is informative. On Llama, attention is intrinsically low-noise (§I.1), so `attn_only` acts almost as a pass-through and lands near `ALL`. On Qwen, `attn.q_proj` carries noise-mass 0.91 (vs. 0.025 on Llama), so leaving the MLP unrepaired collapses the cell. `attn_only` is therefore not a fair control across families: it isolates a different noise source on each.

Across both pairs, `gate_up` is the highest- r mask whose per-benchmark accuracies match `ALL` (retention 0.78 and 0.73 vs. 0.31 and 0.19), capturing the top-two noise classes from §I.1 while preserving 73 to 78 percent of the FT delta: the operational sweet spot of maximum FT preservation with minimum noise leakage.

I.3 Comparison with element-wise merging baselines

We compare DG-Hard against three element-wise merging baselines on two pairs (Qwen3.5-4B + Winogrande, severe FT, and Llama-3.2-3B + MedQA, mild FT): WiSE-FT and Task Arithmetic [13, 31], which are mathematically identical for a single (base, FT) pair and which we report jointly; TIES [15], which keeps the top k entries of $|\Delta|$, sign-elects, and averages; and DARE [14], which randomly drops entries with probability p and rescales surviving entries by $1/(1-p)$. We use “DARE”

to denote the canonical rescaled form (`rescale=True`) and “DARE-Linear” to denote the variant without rescaling (`rescale=False`). Each method is evaluated at two operating points. Part A uses each baseline at its published default (WiSE-FT/Task-Arith $\alpha=0.5$; TIES `keep_ratio= 0.2` with $\lambda=1$; DARE `drop_prob= 0.5` with `rescale=True`; DG-Hard at its parameter-free RMT-optimal threshold). Part B bisects every baseline to the same Frobenius rollback $r=0.5$, isolating shape effects from magnitude effects; this matched-rollback comparison is the central methodological control. For DARE in Part B we use the linear no-rescale form (DARE-Linear), since DARE with `rescale=True` at $p=0.5$ has expected Frobenius ratio $\sqrt{1/(1-p)} \approx 1.41$, which is incompatible with the matched $r=0.5$ target.

Table 20: Merging baselines on Qwen3.5-4B + Winogrande (severe FT damage). $D = \{\text{arc_challenge, gsm8k}\}$, $I = \{\text{hellaswag, ifeval, mmlu}\}$, $U = \emptyset$. Combined uses the protocol harmonic mean over the partition. Winning Combined per part in bold. DARE’s [14] expected Frobenius ratio is $\sqrt{1/(1-p)} \approx 1.41$ at $p=0.5$; the realized $r=1.155$ reflects single-seed mask variance. Combined of 0 indicates a sub-statistic clipped to its floor (e.g., % preserved non-positive on the improved partition), which collapses the harmonic mean.

Method	r	GSM8K	ARC-C	HSwag	IFEval	MMLU	task_win	Combined
<i>Part A: canonical operating points</i>								
WiSE-FT / Task-Arith $\alpha=0.5$	0.500	91.1	81.7	60.9	35.5	72.6	85.7	67.5
TIES canonical	1.000	57.3	62.6	87.1	41.4	75.8	89.8	0.0
DARE canonical	1.155	56.3	65.8	87.0	43.2	75.3	89.6	7.9
DG-Hard canonical	0.314	87.7	73.9	85.0	36.1	75.7	89.4	72.6
<i>Part B: matched Frobenius rollback $r=0.5$</i>								
WiSE-FT / Task-Arith $r=0.5$	0.500	91.1	81.7	60.9	35.5	72.6	85.7	67.5
TIES $r=0.5$	0.504	92.7	87.4	47.3	35.2	65.0	68.0	21.3
DARE-Linear $r=0.5$	0.354	92.4	87.1	47.3	35.4	64.8	79.7	22.0
DG-Hard $r=0.5$	0.500	90.3	75.3	87.1	38.3	79.8	89.3	81.5

Table 21: Merging baselines on Llama-3.2-3B + MedQA (mild FT damage). $D = \{\text{gsm8k}\}$, $I = \{\text{hellaswag, mmlu}\}$, $U = \{\text{arc_challenge, ifeval}\}$. Winning Combined per part in bold. DARE’s [14] expected Frobenius ratio is $\sqrt{1/(1-p)} \approx 1.41$ at $p=0.5$; the realized $r=1.155$ reflects single-seed mask variance. Combined of 0 indicates a sub-statistic clipped to its floor (e.g., % preserved non-positive on the improved partition), which collapses the harmonic mean.

Method	r	GSM8K	ARC-C	HSwag	IFEval	MMLU	task_med	Combined
<i>Part A: canonical operating points</i>								
WiSE-FT / Task-Arith $\alpha=0.5$	0.501	71.1	74.6	67.5	76.9	38.1	58.7	72.8
TIES canonical	1.000	67.2	74.5	70.1	75.2	55.7	58.1	5.4
DARE canonical	1.155	66.8	74.4	70.2	76.0	55.3	57.7	0.0
DG-Hard canonical	0.186	73.8	74.6	67.1	75.9	38.7	57.7	79.5
<i>Part B: matched Frobenius rollback $r=0.5$</i>								
WiSE-FT / Task-Arith $r=0.5$	0.501	71.1	74.6	67.5	76.9	38.1	58.7	72.8
TIES $r=0.5$	0.503	70.4	74.5	64.8	78.3	34.0	57.6	47.4
DARE-Linear $r=0.5$	0.354	70.1	74.8	65.3	78.0	31.3	58.0	42.5
DG-Hard $r=0.5$	0.427	72.8	74.1	68.8	75.9	44.7	57.4	90.4

Canonical operating points. On Qwen + Winogrande, DG-Hard wins on Combined (72.6 vs. 67.5 for WiSE-FT/Task-Arith) and improves the damaged benches relative to TIES/DARE (GSM8K 87.7 vs. 57.3/56.3; ARC-C 73.9 vs. 62.6/65.8). On Llama + MedQA, TIES and DARE post slightly higher MMLU than DG-Hard (55.7/55.3 vs. 38.7), yet their Combined collapses to 5.4 and 0.0 respectively. The explanation is visible in the r column: at canonical settings TIES and DARE sit at $r \approx 1$, so they do not shrink the FT delta and therefore heal a negligible fraction of the FT-induced damage on GSM8K. The partition-conditional Combined headline reveals this gap; an unweighted

per-benchmark mean would not. WiSE-FT and Task Arithmetic produce identical outputs for a single (base, FT) pair, so we report a single row and credit both citations.

Matched Frobenius rollback $r=0.5$. With magnitude held fixed, DG-Hard wins both pairs on Combined (+14.0 pp on Qwen + Winogrande, 81.5 vs. 67.5; +17.6 pp on Llama + MedQA, 90.4 vs. 72.8). The per-benchmark breakdown shows why: TIES at $r=0.5$ collapses HellaSwag (47.3 vs. 87.1 under DG-Hard) and DARE-Linear at $r=0.5$ collapses ARC-C relative to DG-Hard’s distribution of mass while also collapsing HellaSwag (47.3). DG-Hard is the only method that rolls back to $r \approx 0.5$ without driving any of the five benchmarks to a degenerate value. TIES and DARE-Linear sparsify in the standard parameter basis while DG-Hard sparsifies in the singular-vector basis; both bets land at the same Frobenius rollback, but spectral sparsity beats coordinate sparsity at the matched rollback point.